Auditing Correlated Failures in Frozen-Feature Pretrained-Encoder Pools for Medical Segmentation

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Abstract

Medical segmentation pools are often used as if additional members provide additional failure evidence, but that assumption is questionable when members reuse the same frozen pretrained encoder bundle, input statistics, and decoder recipe. We contribute a reusable E&D audit protocol and trace-level evidence bundle for this setting: measure case-level co-failure by per-case Dice-score correlation, separate the Table-1 same/release-recipe group from a prespecified cross-bundle reference, and release the traces needed to recompute the audit. Across five primary task rows (Kvasir polyp, ACDC LV, BraTS foreground, RIGA Cup, RIGA Disc), functional pools of 9–11 pretrained encoder bundles with four decoder seeds yield same/cross Pearson gaps of $\Delta = 0.261$–0.638 after a Dice ≥ 0.30 retrospective functional floor; both case-level and hierarchical bootstrap 95% CIs for Δ exclude zero in this finite audit. The sign of the same/cross gap is stable across estimator and diagnostic checks, including floor sweeps, calibration-selected functional pools, binary failure events, ICC(A,1), ACDC/BraTS subject-level resampling, pair-quality controls, and cross-fitted item-difficulty residualisation. The practical implication is narrower: in this frozen/light-adaptation regime, same frozen-checkpoint/recipe decoder seeds should not be counted as independent failure evidence without an in-domain co-failure audit, and same-family cross-checkpoint bundles should be reported separately rather than assumed independent. Secondary pixel-error, lift, same-architecture-family ViT / different-pretraining, same-family cross-checkpoint, and same-backbone diagnostics describe where dependence is observed in this benchmark, while leaving clinical safety and causal separation of architecture, pretraining, and recipe outside the supported claim. The artifact supports trace-level recomputation from per-case Dice traces, summary JSONs, metadata, hashes, scorer code, and selected appendix summaries allowed by upstream redistribution constraints; it is not raw-data end-to-end reproduction.

1 Introduction

Ensembles are often used in safety-relevant medical segmentation because different members may fail on different cases. Aggregate Dice can tell us whether members are accurate, but it does not by itself show whether additional members provide additional failure evidence on the same cases. This paper audits that missing quantity for frozen or lightly adapted pretrained-encoder segmentation pools: the aligned case-level correlation of segmentation quality across pool members.

The medical-imaging supply chain now reuses a small set of foundation-model encoders and conventional pretrained backbones: DINOv2, CLIP/BiomedCLIP, SAM/MedSAM, ConvNeXt, EfficientNet, ResNet, and domain-specific pretrainings such as RETFound. A pool of "different" segmenters can therefore share backbone family, pretraining bundle, input statistics, and training recipe. This creates an evaluation gap: aggregate Dice and common uncertainty summaries usually do not

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directly report aligned case-level co-failure, which is the evidence needed for independent failure-coverage claims. The intuition that near-duplicate pipelines should share errors is not new; the missing piece is a released, case-aligned audit that quantifies how large the dependence is, tests whether it survives simple alternative explanations, and states which evaluation claims the evidence can and cannot support. We use **regime-specific co-failure dependence** for this measured pattern in a tested frozen/light-adaptation regime, not for every segmentation ensemble, prompt-conditioned MedSAM, full nnU-Net pipeline, or clinical outcome.

Primary contribution. We contribute a reusable E&D audit protocol for fixed segmentation pools: compute per-case Dice traces, define a prespecified Table-1 same/release-recipe group and cross-bundle reference, average symmetrically over decoder seeds, and report paired bootstrap intervals for the same/cross correlation gap. The protocol asks a concrete reporting question: before a pool is credited with complementary failure coverage, have same/release-recipe and cross-bundle co-failure correlations been measured on aligned cases?

Artifact and diagnostic contribution. We release the trace-level files and scorer code needed to recompute the primary audit statistics. Additional diagnostics test estimator sensitivity, subject-level clustering, item difficulty, pixel error overlap, same-family cross-checkpoint structure, same-backbone perturbations, adaptation and full-pipeline boundaries, and selected scope checks; these diagnostics are not used as causal decomposition. The reusable object contributed by this paper is therefore an audit protocol plus a trace-level evidence bundle for measuring whether a fixed pool of segmentation models supplies independent failure evidence on aligned cases.

2 Related work

Relation to prior work and scope of novelty. Prior work has established that correlated errors and monoculture can matter. Our narrower contribution is to make this issue auditable for a concrete medical segmentation setting by releasing aligned per-case traces, same/cross estimators, and a primary/diagnostic evidence hierarchy. Our risk framing follows algorithmic-monoculture and foundation-model ecosystem work [Kleinberg and Raghavan, 2021, Bommasani et al., 2021, 2022, Toups et al., 2023]. The correlation gap we measure has modelling antecedents: trial-by-trial error consistency asks whether systems fail on the same inputs [Geirhos et al., 2020], error correlations arise when deployed models share algorithms, datasets, or foundation models [Li et al., 2025], shared training trajectories and weight-space basins can constrain checkpoint diversity [Fort et al., 2019, Wortsman et al., 2022], transfer/representation work studies what is reused across pretrained models [Neyshabur et al., 2020, Gontijo-Lopes et al., 2022], pretrained-model ensemble work treats pretraining as a possible diversity source [Mustafa et al., 2020], same-architecture models can align strongly across initializations [Kornblith et al., 2019], and predictive-diversity pathologies can limit deep-ensemble gains [Abe et al., 2024]. Recent LLM monoculture and null-model papers are relevant because they show that agreement/correlation claims depend on the choice of baseline and unit of analysis; our paper applies that lesson to case-level medical segmentation traces [Jiang et al., 2025, Kim et al., 2025a, Gorecki and Hardt, 2025, Goel et al., 2025, Jo et al., 2026]. Pathology-FM and precision-oncology studies provide adjacent evidence about center robustness, benchmark behavior, and complementary domain-specialised features [de Jong et al., 2025, Campanella et al., 2025, Luo et al., 2025, Neidlinger et al., 2025]. Our contribution is not the abstract claim that correlated errors exist; rather, we provide a released medical-segmentation audit of frozen-feature pretrained-encoder pools that distinguishes primary from diagnostic evidence and reports where same-checkpoint/recipe, same-architecture-family, and broader cross-family correlations appear or contract under a shared protocol.

Medical segmentation, pretrained/foundation-model benchmarks, and ensemble baselines. nnU-Net, deep ensembles, MC dropout, adjacent uncertainty or parameter-efficient ensembles, and hyperparameter ensembles are usually evaluated through Dice, calibration, or uncertainty quality [Isensee et al., 2021, 2024, Lakshminarayanan et al., 2017, Gal and Ghahramani, 2016, Mehrtash et al., 2020, Jungo and Reyes, 2019, Mühlematter et al., 2026, Gu et al., 2024]. Classical ensemble-diversity metrics motivate this audit, but our reporting target is the case-aligned correlation of task-performance traces because that statistic directly tests redundant failure evidence on the evaluated benchmark [Kuncheva and Whitaker, 2003, Ganaie et al., 2022, Wood et al., 2023]. Medical-segmentation diversity and failure-detection benchmarks ask whether a method can detect segmentation failures; our audit asks whether multiple pool members fail on the same cases, making the questions complementary rather than substitutable [Georgescu et al., 2023, Zenk et al., 2025]. Recent medical-segmentation benchmarks and robustness-specification work evaluate or discuss generic-vision probes, SAM medical-image evaluations, standardized segmentation datasets, promptable SAM adaptations, universal 3D/medical segmenters, task-specific robustness requirements, transferability estimation, and VFM segmentation benchmarking [Baharoon et al., 2023, Huang et al., 2024, Kuş and Aydin, 2024, Kirillov et al., 2023, Ma et al., 2024, Cheng et al., 2023, Archit et al., 2025, Zhu et al., 2024, Ma et al., 2025, Gu et al., 2025, He et al., 2025, Kim et al., 2025b, Xian et al., 2025, Yang et al., 2023, Kerssies et al., 2024]. These benchmarks are complementary to our audit: they ask which model or foundation model performs well, whereas we ask whether multiple members of a pool provide non-redundant failure evidence on the same cases. Promptable and 3D medical foundation models are relevant deployment alternatives, not baselines for this frozen-encoder co-failure audit.

3 Formal setup and benchmark estimands

Pool and estimands. Let $\mathcal{F}=\{F_1,\ldots,F_K\}$ be the operational pretrained-encoder bundles used by the Table 1 scorer. In the primary 11-bundle source pool, a bundle is the concrete released encoder / pretraining / input-statistic wrapper used for frozen-feature extraction; same-bundle decoder seeds form most of the main same group, and DINOv2-B×DINOv2-S is the only recurring cross-checkpoint member of that group because both variants share the same upstream DINOv2 release and pretraining recipe. Other broad-family groupings (for example OpenAI CLIP sizes, ResNet sizes, and expanded DINOv2/ConvNeXt variants) are not folded into the Table 1 primary estimator; they are separated in the released same-family cross-checkpoint diagnostic in Appendix A31. BiomedCLIP (ViT-B/16, biomedical image-text pretraining) is treated as a different bundle from DINOv2 despite being a ViT. A model trace is a tuple (F_k,σ) (bundle, decoder seed) producing per-case Dice $D_{(k,\sigma)}(i,t)$. Appendix A30 and Appendix A31 report released diagnostics that separate Table-1 product bundles, same-architecture-family ViT / different-pretraining, and broader same-family cross-checkpoint readouts; these clarify that a "cross" column can still contain broad architectural structure. For a pair of models $a\neq b$ we define $r_{a,b}(t)=\mathrm{Pearson}_i[D_a(i,t),D_b(i,t)]$ (and a McGraw–Wong ICC(A,1) variant $\iota_{a,b}$ [McGraw and Wong, 1996], chosen to penalise per-bundle Dice offsets that Pearson absorbs). The Table-1 same/cross means at task t are the expectations of $r_{a,b}$ over pairs in the same/release-recipe group (same bundle, plus DINOv2-B/S) and the remaining Table-1 cross reference under this scorer, and $\Delta(t)=\bar{r}_{\mathrm{same/release}}(t)-\bar{r}_{\mathrm{cross}}(t)$ is the co-failure gap. Secondary estimands are pixel-level error-mask Dice, computed the same way over binarised error masks, and joint-failure *lift* over independence $L_{\mathrm{mono}}(\tau)=\Pr[\text{all fail}\mid\text{mono pool}]/\prod_m\Pr[\text{fail}_m\mid\tau]$.

Terminology used by the audit. The primary claim uses only the Table-1 same/release-recipe group and the Table-1 cross reference; broader family rows are diagnostics.

Term	Meaning in this paper	Primary claim?
Bundle	Concrete encoder checkpoint plus wrapper, pretraining, input normalisation, and feature-extraction path	Yes
Model trace	Bundle plus decoder seed and aligned per-case Dice vector	Yes
Table-1 same/release-recipe group	Same-bundle seed pairs plus the prespecified DINOv2-B/S same-release cross-checkpoint pair	Yes
Table-1 cross reference	Remaining primary-pool pairs under the Table-1 grouping	Yes
Same-family cross-checkpoint	Broader DINOv2, CLIP, ConvNeXt, and ResNet diagnostic grid	Diagnostic only
Same-architecture ViT cross-pretraining	ViT-B diagnostic varying pretraining source	Diagnostic only

Symmetric multi-seed averaging. For each bundle pair we enumerate all $|\Sigma|^2=16$ seed combinations (4 per model) rather than seed-matched pairs, removing the one-seed-per-bundle noise that inflates ad-hoc gap estimates. Same-bundle averaging uses $\binom{|\Sigma|}{2}=6$ same-bundle seed pairs per F_k plus the full $|\Sigma|^2$ on DINOv2-B×DINOv2-S; the cross group uses all remaining Table-1 bundle pairs.

CI and scope. A 3-level nested hierarchical bootstrap (Table-1 groups → seeds → cases, $B=2000$) yields 95% CIs as the 2.5th/97.5th percentiles of bootstrap-sampled $\bar{r}_{\mathrm{same}},\bar{r}_{\mathrm{cross}},\Delta$. Because the number of encoder-bundle clusters is small, neither interval should be read as a population-level guarantee over all possible pretrained encoders. We report both intervals as finite-audit uncertainty summaries: the case bootstrap asks about new cases conditional on the evaluated pool, while the hierarchical bootstrap also perturbs the finite group/seed surface. The estimands assume a shared-decoder, shared-task pool and measure *observed* correlation within such pools; a released CNN random-init diagnostic (Appendix A33.2) shows positive same-vs-cross structure without pretrained weights, so we report the pretrained-bundle gap descriptively rather than as a causal isolation of pretrained-weight effects.

Primary-group structural caveat. The primary rows retain task-specific functional pools of 11/10/10/9/11 bundles after the Dice floor. In these pools, the Table 1 "same" column is dominated by same-bundle, different-seed decoder-head pairs, with DINOv2-B×DINOv2-S providing the only recurring same-family cross-checkpoint pair when both variants are functional. Thus the high same-group $\bar{r}$ values in Table 1 (0.834–0.959) are best read as same-frozen-checkpoint/recipe-dominated dependence readouts, not a broad sample of many same-family cross-checkpoint comparisons. Section 5.1 and Appendix A30 report a 5-variant same-architecture-family ViT / different-pretraining diagnostic (DINOv2-B, BiomedCLIP, MAE-B, DeiT-B, CLIP-B) that measures cross-pretraining correlation under a shared broad architecture family and is the closest "within-architecture-family, cross-pretraining" control in this benchmark.

4 Experimental design

Datasets & targets. The primary rows are summarized below. The splits are recorded in each JSON's `test_ids`; subject-level sensitivity for ACDC and BraTS is in Appendix A11.

Primary task rows and evaluation units. The audit uses benchmark-derived targets and aligned per-case Dice traces, not clinical deployment cohorts.

Task row	Raw source	Target / split	Unit	Main caveat
Kvasir polyp	Kvasir-SEG [Jha et al., 2020]	Loader-defined 880/120 image split	image, $N=120$	No patient/video IDs available; no patient-level duplicate control claimed.
ACDC LV	ACDC [Bernard et al., 2018]	LV-positive ED/ES slices from public patients 001–100, fold 0	slice, $N=361$; 20 patients	Slice row; patient-level sensitivity reported in Appendix A11.
BraTS foreground	BraTS 2024 GLI [BraTS 2024 Organizers, 2024, de Verdier et al., 2024]	2D FLAIR `seg>0`; top-10 eligible slices per held-out subject, NumPy seed 42	slice, $N=3{,}228$; 324 subjects	Includes resection-cavity label; not the official 3D BraTS WT challenge target.
RIGA Cup/Disc	RIGA+ [Almazroa et al., 2018, Almazroa, 2018]	BinRushed+MESSIDOR train, Magrabia source test	fundus, $N=95$	Source/subfolder labels retained only for deterministic alignment.

Encoder bundles. *Generic pretrained-encoder bundles* (11, mixing foundation-model encoders and conventional pretrained backbones): DINOv2-B/S, BiomedCLIP, CLIP ViT-L/14, CLIP ViT-B/16, MAE ViT-B/16, DeiT ViT-B/16, ConvNeXt-T, EfficientNet-B0, ResNet-50/18. Encoder names refer to the exact released wrapper/checkpoint paths in the artifact; in particular, CLIP-L/14 is the documented OpenCLIP dense-token wrapper rather than a claim of official OpenAI inference parity. *Medical-domain encoder probe:* MedSAM vision-encoder-only (SAM ViT-B; prompt encoder and mask decoder discarded) is reported as a secondary boundary probe on RIGA/Kvasir/ACDC. RETFound support is documented as a gated-checkpoint path, but RETFound cells are not reported because the checkpoint/authentication was unavailable for the reported artifact run.

Protocol. The primary protocol resizes images and masks to 224×224 with nearest-neighbor mask resizing, applies encoder-specific upstream normalisation (ImageNet-style statistics for DINOv2/MAE/DeiT/torchvision CNNs and OpenAI CLIP statistics for CLIP/BiomedCLIP), freezes the encoder, and trains a 1×1-conv decoder with Adam 10^{-3} for 30 epochs, batch 16, BCE loss on the binary primary rows, no early stopping, no validation-based model selection, and fixed sigmoid threshold 0.5 at inference. Table 1 starts from the 11-bundle generic source pool where cells are available and applies the estimator's Dice≥0.30 retrospective functional floor to remove non-segmenting, near-constant traces: Kvasir retains 11 bundles, ACDC 10, BraTS foreground 10, RIGA Cup 9, and RIGA Disc 11. This floor is a test-trace-based analytic convention chosen for functional-pool definition, not a training filter, ingestion filter, deployment rule, or pre-registered hypothesis threshold; Table 1 is therefore conditional on the filtered functional pool, and Appendix A5 reports no-floor and stricter-floor sensitivity with retained pool counts. The MedSAM sensitivity is a secondary encoder-only boundary probe, not part of the primary pool or a prompt-conditioned MedSAM evaluation.

5 Main result: same-frozen-checkpoint pools show higher co-failure correlation

Table 1: **Primary finite-pool co-failure audit.** For each filtered functional pool, we compute average pairwise Pearson correlation of per-case Dice traces for the Table-1 same/release-recipe group and the Table-1 cross reference. K is the retained bundle count after the Dice≥ 0.30 functional floor; M is the number of retained model traces; pair counts are same/cross model-trace pairs. CIs are 95% bootstrap intervals for $\Delta = \rho_{\text{same/release}} - \rho_{\text{cross}}$. ACDC and BraTS are slice-level rows; subject-level sensitivity is reported in Appendix A11.

Task row	Unit N	K	M	pairs S/C	$\rho_{\text{same/release}}$	ρ_{cross}	Δ	Δ case CI[§]	Δ hier CI[¶]
Kvasir polyp	image 120	11	44	82/864	0.958	0.697	0.261	[0.211,0.320]	[0.174,0.378]
ACDC LV	slice 361[‡]	10	40	76/704	0.959	0.639	0.321	[0.289,0.355]	[0.211,0.435]
BraTS foreground[†]	slice 3,228[⋆]	10	40	76/704	0.938	0.623	0.315	[0.302,0.329]	[0.224,0.425]
RIGA Cup	fundus 95	9	36	70/560	0.834	0.361	0.473	[0.395,0.556]	[0.280,0.780]
RIGA Disc	fundus 95	11	44	82/864	0.870	0.232	0.638	[0.569,0.687]	[0.475,0.831]

[§] Case-level paired bootstrap from `gap_ci95_case` (B=2,000), conditional on the functional pool shown. [¶] Hierarchical bootstrap over Table-1 grouping and seed clusters from `gap_ci95_hier`; it is a finite-pool interval, and all hierarchical CIs exclude zero. [‡]20 ACDC patients. [⋆]324 BraTS subjects. [†]BraTS uses the completed 11-bundle source pool for the 2D FLAIR `seg>0` derivative; ResNet-18 is below the Dice floor, leaving 10 retained bundles.

Across four imaging domains and five task rows, the filtered functional pools contain 36–44 model traces. Within these finite pools, the same/release-recipe group has higher average per-case Dice correlation than the Table-1 cross reference on every row ($\Delta = 0.261$–0.638). The CIs in Table 1 summarize uncertainty for Δ, not a population-level claim over all possible encoders. Table 2 summarizes four sign-stability diagnostics: a calibration-selected functional pool evaluated on disjoint cases, a binary failure-event readout at the empirical P_{33} threshold, quality-controlled pair regression, and distinct-checkpoint same-family decomposition.

Table 2: **Sign-stability diagnostics for the primary rows.** Values are point estimates from distinct diagnostics and are not directly comparable in scale. Cal.-split Δ selects functional bundles on a deterministic calibration subset and evaluates the same/cross gap on disjoint cases. P_{33} fail ϕ gap is the same/cross gap in binary failure-event correlation at the empirical 33rd percentile of model-case Dice after functional-floor filtering. Quality-control β_{same} is the same-group coefficient in a pair-level OLS diagnostic with case-bootstrap refits. Expanded-grid xckpt–cross is the same-family cross-checkpoint diagnostic from Appendix A31, not the Table 1 analytic-pool cross-family stratum. Full definitions and CIs for the Table 2 columns are in Appendices A5–A7 and Appendix A31; the separate item-difficulty diagnostic is in Appendix A27.

Task	Cal.-split Δ	P_{33} fail ϕ gap	QC β_{same}	Xckpt–cross
Kvasir polyp	0.250	0.329	0.169	0.080
ACDC LV	0.350	0.386	0.208	0.134
BraTS foreground	0.307	0.413	0.224	0.093
RIGA Cup	0.478	0.453	0.372	0.267
RIGA Disc	0.630	0.578	0.582	0.227

Estimator robustness. A released CPU-only sensitivity file (`results/_merged/diagnostics/audit_sensitivity.json`) recomputes the primary Pearson gap under a functional-floor sweep, an ICC(A,1) alternative, leave-one-bundle/task-out summaries, and a random family-label permutation stress test. Analytic-floor checks show the gap remains positive at no-floor and stricter-floor settings where the pool is non-empty. Event and ICC checks show the same sign after binarising failures at the empirical P_{33} threshold and after replacing Pearson with ICC(A,1). Quality and residualisation checks show that simple marginal-quality controls and held-out item-difficulty residualisation do not absorb the same/cross separation. In 1000 random-label permutations the observed row gap is exceeded with rank fraction 0.001 on all rows; because family labels are not exchangeable sampling units, we treat this as a label-randomisation stress test, not a population-level p-value.

Spatial and lift readouts. Figure 1 is not used as primary evidence for the Table 1 claim. It illustrates why continuous case-level correlation is worth auditing: in selected cases, same-bundle seeds can produce spatially overlapping false positives and false negatives, and secondary aggregate pixel/lift diagnostics show the same qualitative dependence pattern. In the released aggregate diagnostics, same-family pairwise error-mask Dice is 0.80–0.87 versus 0.32–0.40 cross-family on

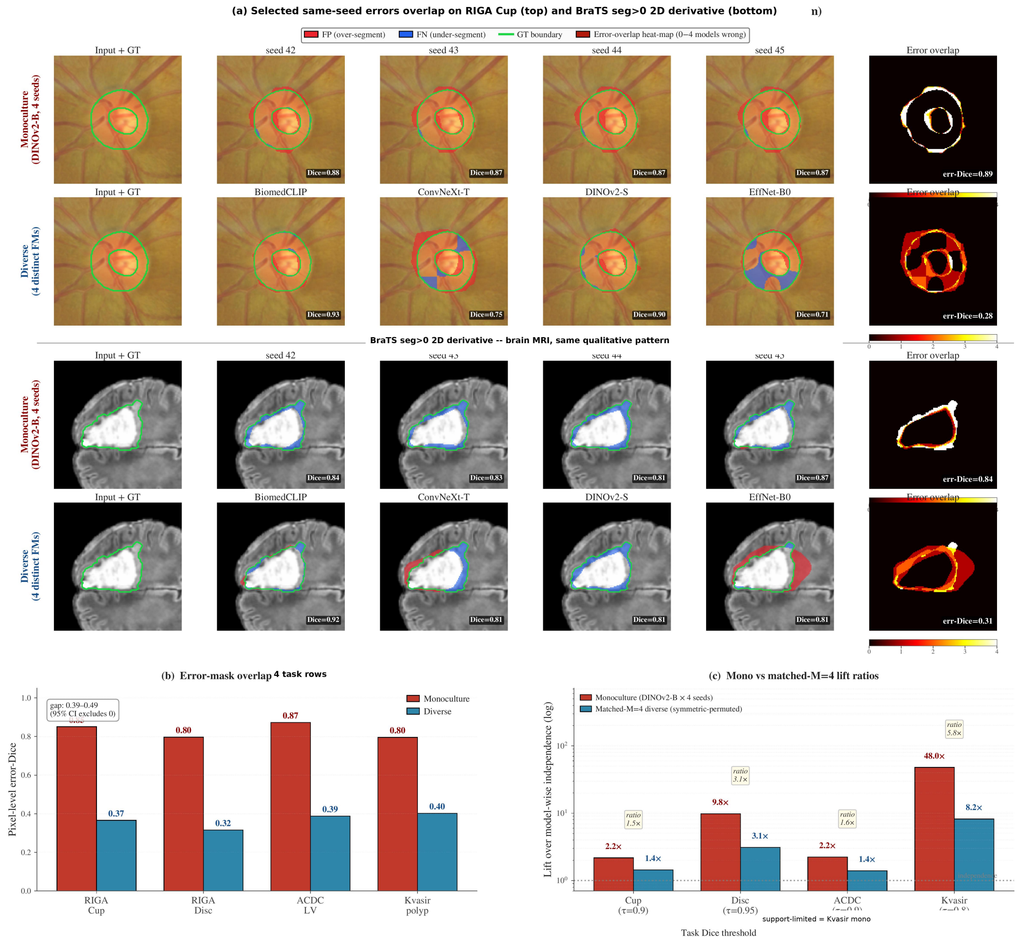

Figure 1: **Secondary spatial and threshold diagnostics illustrating the co-failure audit.** Table 1 is the primary case-aligned per-case Dice evidence. **(a)** Selected illustrative cases compare four DINOv2-B decoder seeds with four distinct-bundle members on RIGA Cup and the BraTS 2D foreground derivative (`seg>0`; not the official 3D BraTS WT challenge target). Red denotes false positives, blue false negatives, and green the ground-truth boundary; cases were chosen as visual examples where same-bundle and diverse pools have comparable mean Dice but visibly different error overlap. **(b)** Aggregate pixel-level error-mask Dice on secondary diagnostic splits; these rows are not the Table 1 primary case-aligned rows. **(c)** Matched-$M=4$ joint-failure lift diagnostics with support flags; support-limited rows are descriptive and not headline evidence. The artifact releases aggregate summaries for panels (b–c), not raw masks or full prediction caches.

RIGA Cup/Disc ($N=224$), ACDC LV ($N=90$), and Kvasir ($N=300$) rows (Fig. 1b, Appendix A3, and Appendix A8). Because these diagnostics use secondary splits or aggregate-only artifacts, we interpret them as explanatory companions to Table 1; sparse rare-event rows are descriptive rather than headline statistics.

Item-difficulty diagnostic. Following the warning that co-failure claims depend on the null model [Jo et al., 2026], we residualise per-case Dice against held-out item-difficulty estimates. The naive all-bundle difficulty regressor leaks the response and inflates Δ. A released all-row cross-fitted diagnostic estimates each trace's item-difficulty regressor from traces outside that trace's broad upstream family, then recomputes the Table-1 same/cross rule on residuals. Because residualisation changes the correlation scale and can increase same/cross gaps, we interpret this diagnostic only qualitatively: held-out item-difficulty residualisation does not absorb the same/cross separation on any primary row (Appendix A27).

5.1 Limited ViT diagnostic: cross-pretraining correlations remain intermediate

A natural Kornblith-type concern [Kornblith et al., 2019] is that two ViTs produce similar representations *because they share architecture family*, not because they share pretraining. We hold broad architecture family fixed in a limited 5-variant ViT pool—DINOv2-B (SSL on LVD-142M), BiomedCLIP (CLIP on biomed image-text), MAE-B (ImageNet MAE), DeiT-B (ImageNet supervised), OpenAI CLIP-B—trained with the identical frozen 1×1-conv probe (Table 3). The released aggregate covers RIGA Cup and ACDC LV. In this limited diagnostic, ViT cross-pretraining correlations lie between the same-bundle seed floor and the primary cross-family reference; this ordering is descriptive only and does not identify which factor is causal.

Table 3: **Limited ViT same-architecture/different-pretraining diagnostic.** The table compares same-bundle seed-pair correlations with cross-pretraining correlations among ViT-style encoders. It is a two-task diagnostic, not a causal decomposition of architecture and pretraining.

Dataset	ViT subset	Func. ViTs	Seed floor	ViT cross-pretrain	Primary cross ref.	Seed floor − ViT cross-pretrain
RIGA Cup	all 5 ViTs	5/5	0.968	0.680	0.361	0.288
ACDC LV	CLIP-B dropped†	4/5	0.988	0.807	0.639	0.182

Values are from `results/_merged/diagnostics/r7_arch_confound_analysis.json` and `paper_table.json`.
†The all-5 ACDC same-architecture-family ViT cross-pretraining value is 0.631 because CLIP-B is nonfunctional on ACDC (mean Dice 0.205); the functional-filtered readout is 0.807. ViT$\times$CNN subcells are kept as appendix diagnostics and are not used as main-table numeric evidence.

Because the primary same column is dominated by same-bundle decoder-seed pairs, we also release a CPU-only same-family cross-checkpoint decomposition over the five primary rows (Appendix A31). It separates the same-bundle seed floor, distinct checkpoints within broad upstream families (DINOv2-S/B, OpenAI CLIP-B/L, ResNet-18/50 when functional), and cross-family pairs. The distinct-checkpoint same-family minus cross-family gap remains positive on all five rows (0.080–0.267), while staying well below the same-bundle seed floor. Thus the primary effect is strongest for shared checkpoints and recipes, but the released traces do not collapse to a same-decoder-seed artifact.

6 Same-backbone perturbation diagnostics and mechanism limits

(C1) Head architecture/capacity. The released head-capacity and UNet-skip decoder perturbation diagnostics change accuracy and reduce some pairwise correlations, but their same-backbone correlations remain above the primary RIGA cross-family reference. Thus the tested decoder-capacity changes did not remove same-backbone co-failure in these released diagnostics.

(C2) Stochastic/adaptation levers. In the released RIGA Cup diagnostic, MC averaging modestly lowers within-bundle correlation but remains far above the cross-family reference. Bootstrap-bagging and fold-reslicing are appendix diagnostics with different sampling surfaces; the fold-reslicing row is a single-seed split-stability check, not a within-fold versus cross-fold same-backbone estimator. Full-fine-tuning and LoRA probes whose raw caches are not distributed are kept only as transparency notes and are not used as artifact-supported evidence.

(C3) Medical-domain encoder-only probe. A MedSAM image-encoder-only sensitivity is described in Appendix A10 and released as an aggregate diagnostic summary rather than primary evidence. The probe strips MedSAM's prompt encoder and mask decoder, so it is not a fair evaluation of prompt-conditioned MedSAM as released. RETFound is documented only as a gated-checkpoint path; without the gated checkpoint/authentication, we do not report RETFound rows or use them to support the claims.

7 Evidence taxonomy and scope boundaries

Same-backbone perturbations are useful but incomplete in our measurements. Data bootstrap-bagging reduces within-pool $\bar{r}$ by only 0.05–0.12, the released fold-reslicing diagnostic keeps cross-bundle $\bar{r}$ stable across held-out folds but has only one seed per fold, and the released MC-dropout diagnostic remains a high-correlation same-bundle readout (Appendices A14, A17, A33.7). Released and explicitly labelled appendix diagnostics suggest that shared pretrained weights are not the only

Table 4: **Same-backbone perturbation diagnostics.** These comparisons vary decoder capacity, skip structure, or MC dropout under limited grids. They test whether simple same-backbone perturbations remove high correlation; they are not dose-matched causal controls.

Diagnostic	Released source	Main readout	Cautious interpretation
Head-capacity sweep, RIGA Cup	Released trace tree, App. A33.1	4 heads, 769–3.94M params; Dice 0.725–0.848; cross-head same-bundle $\bar{r}$=0.690 (range 0.510–0.896)	Head design changes accuracy and some pairwise r, but stays above primary RIGA cross-family $\bar{r}=0.361$.
UNet-skip decoder replication	Released summary, App. A16	20 cells; Dice 0.806–0.902; seed-pair r=0.774–0.979	Stronger skip head improves accuracy but does not make same-backbone pools look cross-family.
MC dropout, DINOv2-B/RIGA Cup	Released trace tree, App. A33.7	MC $\bar{r}$=0.947, 0.925, 0.893 for p=0.1, 0.2, 0.4	Weak decorrelation lever; largest tested drop is about 0.10.

source of dependence, while the same-architecture-family ViT and CNN random-init readouts indicate that pretraining, architecture, and shared recipe effects remain entangled in this benchmark rather than collapsing to a single identified mechanism (Appendices A33.2, A30).

The strongest caution is the full-pipeline boundary. The nnU-Net v2 2D/3D complements are seeded task-level runs using custom trainer subclasses and task-level splits; the RIGA rows are not case-identical to the Magrabia-source held-out frozen-encoder row. They are descriptive scope checks rather than official nnU-Net benchmark claims, case-identical baselines, or causal decompositions. Appendix A25 reports a 3D segmentation-architecture pilot as appendix scope context only, not as primary evidence.

Table 5: **Evidence hierarchy used in this paper.** Primary claims are supported only by the first row. Other rows are diagnostic, robustness, or limited-scope evidence and should not be read as causal or deployment evidence.

Evidence role	Bundle support	Interpretation
Primary frozen-encoder audit	Per-case Dice, `paper_table.json`, `within_cross/*.json`	Primary main claim: five task rows give $\Delta = 0.261$–0.638 for frozen/light-adaptation 2D pretrained-encoder pools under the Table-1 grouping.
Estimator and event robustness	`audit_sensitivity.json`; `quality_controlled_pairs.json`; `failure_event_summary.json`; `item_difficulty_robustness.json`	Stability checks only: floor/ICC/leave-one-out/permutation, pair-quality controls, binary failure-event readouts, and cross-fitted item-difficulty residualisation; no new training evidence.
Architecture/pretraining and cross-checkpoint diagnostics	`r7_arch_confound_analysis.json`; `cross_checkpoint_family.json`	Same-architecture-family ViT / different-pretraining and same-family cross-checkpoint rows lie between seed floor and primary cross-family reference; non-distributed cross-architecture probes are labelled separately and not used as support.
Same-backbone levers	Released head-capacity, UNet-skip, and MC-dropout traces plus selected aggregate diagnostic summaries; non-distributed probes are explicitly labelled and not used as support	These weaken simple alternatives but are heterogeneous and not dose-matched causal effects.
Full-pipeline complements	Released nnU-Net JSONs; 3D segmentation-architecture pilot reported only as appendix scope context	Boundary evidence: nnU-Net complements give much smaller gaps, while the 3D pilot is not included in the supplementary artifact; do not generalize the 2D frozen-encoder result to all segmentation ensembles.
Held-out stress test	Aggregate held-out JSONs only; no case CIs for the held-out aggregator	External-validity diagnostic: the 11-bundle stress test gives large direction-positive Hippocampus/ISIC rows, a tiny-N positive Prostate row, a near-zero Spleen row, and a Heart floor failure; held-out rows remain secondary scope evidence.

8 Discussion

Practical reading. Report M_{eff} as a threshold-specific dependence readout before treating same-bundle seeds as independent evidence, not as a standalone pool-selection objective. In our released rows, diverse M=4 pools yield M_{eff} from 2.05 (Kvasir) to 2.97 (RIGA Disc) under the paper's equicorrelation readout at $\tau = 0.85$, rather than 4; same-bundle seed pools yield M_{eff} near 1 under the same readout (Appendix A28). The readout should always be paired with marginal Dice and matched-quality comparisons before any pool-selection decision.

Limitations. The evidence is retrospective and benchmark-based, and should be read at the level of the audit it supports. This study does not validate any model or pool for clinical deployment; it audits a reporting assumption about failure evidence on retrospective benchmarks. We do not measure clinician behavior, diagnoses, patient outcomes, prompt-conditioned MedSAM, broad cross-demographic shifts, fairness, or case-identical full-pipeline baselines for every primary row. The evaluated datasets are not claimed to represent all clinical populations, scanners, demographic groups, or deployment settings. The BraTS foreground row is a 2D derivative of the 2024 GLI release rather than the official 3D BraTS challenge target, and all primary rows are finite 11-bundle-panel audits after a task-specific functional floor applied to the evaluation traces. This is not a VFM performance benchmark, SOTA segmentation benchmark, or full fine-tuning study; the frozen/light-adaptation design is chosen to isolate the co-failure reporting problem under a controlled readout. Table 1 is the confirmatory evidence surface; secondary threshold, lift, pool-selection, and held-out rows are exploratory diagnostics rather than multiplicity-adjusted confirmatory tests. CKA and the random-effects representation are descriptive tools, not causal proofs.

What the evaluation supports. The main E&D takeaway is a reporting requirement: when a medical segmentation pool reuses pretrained encoder checkpoints, input statistics, and decoder recipes, aggregate Dice should be accompanied by aligned case-level co-failure correlations before the pool is credited with independent failure coverage. Our released traces show this requirement is not cosmetic: in five primary frozen/light-adaptation rows, same/release-recipe correlations are substantially higher than the prespecified cross-bundle reference. The result is a finite audit of evaluation evidence, not a clinical safety claim or causal decomposition of pretraining and architecture.

Release. The anonymized artifact is available through the supplementary submission and anonymous code URL `https://anonymous.4open.science/r/NIPS_A_ACF-86CD`. It includes primary per-case Dice JSONs, summary tables, code, metadata/manifests, hashes, environment files, a data card for the derived trace resource, selected appendix summary JSONs, and Croissant metadata including the required Responsible AI metadata fields for the derived trace/code artifact. It does not introduce a new raw dataset; the released JSONs are derived trace artifacts that retain upstream de-identified alignment identifiers and source labels, including subject/patient/slice IDs and RIGA source labels, solely for deterministic alignment and subject-level resampling. Released IDs are not intended for individual-level analysis, subgroup profiling, re-identification, inference of protected attributes, clinical diagnoses, or patient outcomes. The bundle does not include raw medical images, raw masks, direct identifiers, full prediction caches, or held-out raw prediction arrays; raw data and checkpoints remain governed by upstream licenses and DUAs.

9 Artifact and reproducibility scope

The artifact supports trace-level recomputation of the numerical claims from fixed derived traces without including raw medical images, raw masks, or prediction caches. Appendix Table 33 maps each class of main-text evidence to its released files and recomputation path. The mapping is intentionally conservative: primary claims are reproducible from per-case Dice traces, while mechanism, pixel, lift, held-out, and full-pipeline rows are labelled as diagnostics or limited-scope evidence when their raw masks, prediction arrays, or case-identical comparators are not included. This separation is part of the E&D contribution: the artifact makes the evidentiary status of each result explicit without requiring readers to reconstruct it from appendix provenance.

The artifact verification script recomputes the primary scorer and CPU-only appendix scorers used by the main-text robustness checks. The SHA-256 manifest covers the released files, and the metadata describes the trace/code artifact rather than a standalone raw dataset. The artifact is trace-level reproducible, not raw-data end-to-end reproducible: users who want retraining from raw medical images must separately obtain upstream datasets and checkpoints under their original access terms.

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A1 Appendix roadmap and evidence inventory

The appendix is organised by evidentiary role rather than by experiment chronology. It starts with companion readouts, then documents primary estimators, uncertainty, task-level sources, and estimator sensitivity. Middle sections collect mechanism diagnostics and same-backbone perturbation analyses; later sections cover adaptation, data swaps, segmentation-pipeline variants, distribution shift, operational metrics, held-out stress tests, and M_{eff}. The final sections record artifact metadata and additional scope analyses. Secondary rows are retained only when they qualify or explain a paper claim, and their captions state when they use non-primary pools or splits. Unless explicitly marked primary or sourced from the primary `results/_merged` summaries, appendix values are diagnostic rather than primary support for Table 1.

CLIP-L notation. Throughout the appendix, CLIP-L/14 refers to the preserved OpenCLIP `ViT-L-14/openai` dense-token wrapper used to generate the released traces. It is not a claim of OpenAI `clip.load` parity; a strict QuickGELU-parity CLIP-L loader would be a separate sensitivity with a distinct FM identifier.

Table 6: **Appendix guide.** Each block has a single role in the argument; cross-references point to the detailed tables rather than repeating them in the main paper.

Block	Evidence role
Estimators and provenance	Case/hierarchical bootstrap, subject clustering, quality-controlled pair regression, functional-floor/ICC/permutation sensitivity, pixel overlap, annotation-source note, and primary dataset details.
Mechanism diagnostics	Same-architecture-family ViT / different-pretraining, CKA, random-effects, and item-difficulty null analyses.
Same-backbone perturbation analyses	Matched-size/quality-gated pool checks, decoder capacity, UNet skips, fixed-backbone folds, MC dropout, released CNN random-init, and labelled scope-limited probes.
Scope boundaries	nnU-Net complements, 3D pilot, BraTS sparse-target sensitivities, loss/augmentation sensitivity, distribution shift, and held-out failures.
Operational metrics	Conditional recovery, matched-M lift, lift ceiling, pool-selection probes, and M_{eff}.
Artifact documentation	Bundle manifest, artifact metadata, requirements, and trace inventory.

A2 Companion readouts and scope

This section records companion readouts that clarify scope without adding new headline claims.

CKA. We computed CKA only as a descriptive feature-similarity readout. The saved partial-regression coefficient artifact is not included in the supplementary artifact, so we do not use partial-regression p-values or coefficients as formal evidence. The main paper relies on pixel-level failure overlap and per-case Dice correlation for mechanism-adjacent interpretation, and treats CKA as a qualitative companion variable rather than as a standalone causal mediator.

RIGA annotation source. Table 1 uses the six-rater majority consensus as ground truth for both RIGA Cup and RIGA Disc, matching the released RIGA per-case Dice traces. RIGA also distributes individual ophthalmologist masks, but individual-rater replay artifacts are not included in the supplementary artifact. The consensus Cup and Disc rows are therefore the only RIGA annotation-source evidence used by the manuscript.

Probability-map diagnostics. The artifact releases per-case Dice traces and selected aggregate pixel/lift diagnostics, but not raw probability maps. We therefore do not use mean-confidence, probability-calibration, or failure-flagging AUROC values as evidence for any paper claim.

A3 Pixel-level error-overlap breakdown

Table 7: **Pixel-level error-overlap diagnostic on selected secondary rows.** Values are same/cross gaps in error-mask overlap on aggregate diagnostic splits, not primary case-level evidence. For error-Dice, no bootstrap replicate out of B=500 produced a non-positive gap on any task cell, giving the finite-B bound $\Pr[\text{gap}\leq 0] < 1/501$; FP/FN/boundary rows are point-estimate breakdowns.

Metric	RIGA Cup	RIGA Disc	ACDC LV
Error-Dice gap	0.485	0.481	0.485
Error-Jaccard gap	0.515	0.481	0.521
FP-only error-Dice gap	0.453	0.494	0.566
FN-only error-Dice gap	0.527	0.464	0.449
Boundary-band (5px)	0.430	0.439	0.433

FP = (pred foreground, GT background); FN = (pred background, GT foreground); boundary-band restricts error masks to pixels within 5 pixels of the GT edge (trimap). All metrics are pair-wise Dice between the two models' error masks (or empty-mask-excluded Jaccard). On RIGA Cup, RIGA Disc, and ACDC LV binary, the similar-magnitude gaps across FP, FN, and boundary variants indicate the spatial co-failure is not driven by a single error type on those three tasks (Kvasir and BraTS not tabulated in this split).

A4 Bootstrap sensitivity of the gap

Table 1 reports 95% CIs from a three-level nested hierarchical bootstrap that resamples family labels, seeds within each family, and cases (B=2000). We additionally compute a *case-only* bootstrap that fixes the family/seed structure and resamples the N test cases with replacement (B=2000). The two procedures target different sampling dimensions: the hierarchical CI jointly perturbs the finite FM/seed/case surface, whereas the case-only CI asks "what if we had a different test set" conditional on the evaluated pool. We report both as sensitivity evidence for the same finite audit rather than as population-level guarantees.

Table 8: **Case-level bootstrap sensitivity.** Regenerated primary-row point estimates and gap CIs from the case-level bootstrap (B=2,000, fixed family/seed aggregation). BraTS uses slice-level resampling; subject-level clustering is reported separately in Appendix A11.

Task	Functional pool	Within	Cross	Gap [95% case CI]
Kvasir	11-bundle	0.958	0.697	0.261 [0.211, 0.320]
ACDC LV	10-bundle	0.959	0.639	0.321 [0.289, 0.355]
BraTS foreground†	10-bundle	0.938	0.623	0.315 [0.302, 0.329]
RIGA Cup	9-bundle	0.834	0.361	0.473 [0.395, 0.556]
RIGA Disc	11-bundle	0.870	0.232	0.638 [0.569, 0.687]

† Completed 11-bundle source pool with 10 retained bundles after functional-floor filtering; see Appendix A11 for slice-vs-subject resampling caveats.

BraTS foreground pool reconciliation. The headline BraTS foreground result is the 10-bundle Table 1 value of Δ=0.315 [0.302, 0.329] (case-level CI), from the completed 11-bundle source pool with 10 retained bundles after functional-floor filtering (BiomedCLIP, CLIP-B/16, CLIP-L/14, ConvNeXt-T, DeiT-B/16, DINOv2-B/S, EfficientNet-B0, MAE-B/16, ResNet-50; ResNet-18 is below the Dice floor). Source: `results/_merged/paper_table.json`. Subset diagnostics whose exact aggregate files are not included in the supplementary artifact are omitted from the released numeric evidence. The headline reported throughout the main text and abstract is therefore the Table 1 value.

All gap CIs strictly exclude zero under the case-level bootstrap, matching the family/seed-level CIs in the main text.

A5 Estimator-robustness and negative-control checks

The primary row uses Pearson correlation and a Dice$\geq$0.30 per-FM functional floor. To make this choice auditable, the released bundle includes a CPU-only sensitivity artifact, results/_merged/diagnostics/audit_sensitivity.json, generated from the same per-case Dice JSONs by code/scripts/compute_audit_sensitivity.py. It recomputes the within–cross gap under a floor sweep, an ICC(A,1) alternative, leave-one-FM-out summaries, leave-one-task-out meta summaries, and a random family-label permutation control.

Table 9: **Estimator sensitivity on primary traces.** Pearson values are the Table 1 point estimates at the Dice$\geq$0.30 floor. ICC(A,1) is an alternate agreement statistic on the same retained members. The floor range reports the minimum/maximum Pearson gap over $\{0, 0.20, 0.25, 0.30, 0.35, 0.40\}$ where the pool is non-empty. The permutation note reports a non-exchangeable random family-label stress test over completed members, using $(1 + \#\{\Delta_{\text{null}} \geq \Delta_{\text{obs}}\})/(1001)$; this is not interpreted as a p-value. Retained counts at every floor are in audit_sensitivity.json.

Task	Members at 0.30	Pearson gap	ICC(A,1) gap	Floor-sweep gap range
Kvasir	44	0.261	0.333	0.261–0.261
ACDC LV	40	0.321	0.401	0.242–0.355
BraTS foreground	40	0.315	0.404	0.315–0.362
RIGA Cup	36	0.473	0.539	0.384–0.529
RIGA Disc	44	0.638	0.716	0.638–0.638

All five rows have observed-gap-exceeded rank fraction 0.001 at the 0.30 floor.

The five-task mean Pearson gap at the paper floor is 0.401 (median 0.321, range 0.261–0.638). Leave-one-task-out means stay in 0.342–0.437; the smallest value occurs when the largest-gap RIGA Disc row is held out. These checks do not add new cases or new model training. They instead reduce the risk that the headline sign is an artifact of the exact functional floor, the Pearson statistic, or arbitrary family labels.

Calibration-selected functional pools. The primary table defines a functional pool using all released evaluation traces. To check whether the sign depends on this convention, the scorer also makes a deterministic split of each task's aligned units, selects bundles by mean Dice on the calibration subset, and evaluates the same/cross gap on disjoint cases. This is a leakage-sensitivity analysis rather than a prospective validation protocol, but it addresses the specific concern that the functional-floor sign is induced by evaluating the floor and the gap on exactly the same cases.

Table 10: **Calibration-selected functional-pool sensitivity.** Functional bundles are selected on the calibration subset and evaluated on disjoint cases under the Table 1 same/cross grouping. Unit-agg. Δ is reported only for nested slice datasets after patient/subject aggregation. ACDC can select 11 bundles here because selection is calibration-only, whereas Table 1 applies the retrospective all-evaluation functional floor. CIs are case-bootstrap intervals on the evaluation subset. Source: results/_merged/calibration_split/*.json, regenerated by code/scripts/compute_all.py.

Task	Cal./eval units	Selected bundles	Eval. Δ [95% CI]	Unit-agg. Δ
Kvasir	42/78 images	11	0.250 [0.196, 0.322]	–
ACDC LV	7/13 patients	11	0.350 [0.313, 0.392]	0.206
BraTS foreground	113/211 subjects	10	0.307 [0.292, 0.323]	0.205
RIGA Cup	33/62 images	9	0.478 [0.369, 0.599]	–
RIGA Disc	33/62 images	11	0.630 [0.534, 0.695]	–

A6 Quality-controlled pairwise-correlation diagnostic

A possible alternative explanation is that same-FM pairs have higher correlation merely because they have matched marginal quality. We therefore add a CPU-only diagnostic that uses the same released per-case Dice traces retained by Table 1 and fits a pair-level OLS model:

$$r_{ab} \sim I_{\text{same}} + \bar{D}_{ab} + |\Delta \bar{D}_{ab}| + \bar{V}_{ab} + |\Delta V_{ab}|.$$

Here $\bar{D}_{ab}$ and $\bar{V}_{ab}$ are the pair mean Dice and average per-case Dice variance, with absolute pair differences as additional controls. The same-group indicator is exactly the primary grouping rule: same FM plus the DINOv2-B/S pair. The coefficient is not a causal estimand, but it checks that the main sign is not removed by simple pair-quality and variance controls. The diagnostic is generated by code/scripts/compute_quality_controlled_pairs.py and released as results/_merged/diagnostics/quality_controlled_pairs.json.

Table 11: **Pair-quality controlled diagnostic.** β_{same} is the coefficient on the Table-1 same-group indicator in a pair-level OLS check. Same-group coefficients remain positive after controlling for pair mean Dice, absolute mean-Dice difference, average per-case variance, and absolute variance difference. CIs are case-bootstrap intervals from B=1000 refits and should be read as diagnostic, not causal, uncertainty.

Task	Members	Same pairs	Cross pairs	β_{same} [95% CI]
Kvasir	44	82	864	0.169 [0.124, 0.220]
ACDC LV	40	76	704	0.208 [0.175, 0.242]
BraTS foreground	40	76	704	0.224 [0.210, 0.239]
RIGA Cup	36	70	560	0.372 [0.297, 0.456]
RIGA Disc	44	82	864	0.582 [0.495, 0.639]

A7 Failure threshold sensitivity

The primary estimand is continuous per-case Dice correlation. To make the co-failure interpretation less dependent on continuous-score correlation, the released artifact also recomputes the Table 1 same/cross grouping on binary failure events. Table 12 reports the empirical P_{33} readout across all five primary task rows; the released JSON also contains P_{25} and P_{50} thresholds. The RIGA Cup all-fail table below is retained as a pool-level threshold diagnostic, and absolute-threshold lift with support flags is in Appendix A9.

Table 12: **Binary failure-event diagnostic at the empirical P_{33} threshold.** Failure is defined as per-case Dice below the 33rd percentile of all model-case Dice values after functional-floor filtering. ϕ is binary-event Pearson correlation averaged over same-group or Table-1 cross-group pairs. Joint fail same/cross is the pairwise joint-failure probability averaged over the corresponding model-trace pairs. CIs are case-bootstrap intervals for the same–cross ϕ gap. This is not a clinically chosen threshold. Source: `results/_merged/diagnostics/failure_event_summary.json`, regenerated by `code/scripts/compute_failure_event_summary.py`.

Task	τ_{P33}	Same ϕ	Cross ϕ	Gap [95% CI]	Joint fail same/cross
Kvasir	0.659	0.869	0.540	0.329 [0.278, 0.388]	27.9/21.6%
ACDC LV	0.548	0.875	0.489	0.386 [0.354, 0.418]	27.7/19.9%
BraTS foreground	0.636	0.838	0.425	0.413 [0.401, 0.425]	27.3/18.7%
RIGA Cup	0.575	0.706	0.253	0.453 [0.399, 0.509]	24.4/14.4%
RIGA Disc	0.857	0.703	0.125	0.578 [0.511, 0.631]	24.8/12.0%

Full τ-sweep of conditional recovery. Table 13 tabulates grand-mean conditional recovery rates $P[\text{pool-mean Dice} \geq \tau \mid \text{any member Dice} < \tau]$ for same-bundle pools (4 seeds of one FM, averaged over all FMs passing the per-task Dice ≥ 0.30 functional floor) and diverse pools (all $\binom{K}{4}$ functional-subset compositions and all released seed tuples) at $\tau \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$. Source: `results/_merged/diagnostics/tau_sweep.json`, regenerated by `code/scripts/compute_tau_sweep.py`. Recovery is τ-fragile in both regimes; the diverse − same-bundle gap (column Δ) is non-monotone in τ, so the table motivates auditing threshold choices rather than claiming monotonic operational superiority of diverse pools.

A8 Kvasir secondary split diagnostic

Pixel-level diagnostic on this split: mono 0.80 vs. diverse 0.40, gap 0.394 [95% CI 0.38, 0.40]. Absolute-threshold lift at Dice<0.7: mono 252× vs. diverse 120× (ratio 2.1×); at Dice<0.8: 48.4× vs. 14.6× (3.3×); at Dice<0.9: 4.6× vs. 1.8× (2.5×). The numbers are slightly smaller than RIGA/ACDC, likely reflecting the larger test set (300 cases vs. 224 / 90) and more variable polyp presentations, but qualitatively identical: mono still shows substantially higher correlated-failure lift than diverse.

A9 Matched-M lift comparison

To ensure the mono vs. diverse comparison is not driven by pool size, we construct an *oracle* matched-M=4 diverse pool by selecting the top-4 bundles by *test-set seed-averaged* mean Dice

Table 13: **Secondary τ-sweep of conditional recovery rate P[ens pass | any fail].** Grand-mean across all functional 4-of-K diverse compositions and all functional same-bundle baselines. Recovery gaps vary with threshold and can become near-zero or negative near ceiling; these values are operational diagnostics, not primary support for the same/cross correlation claim.

Task	τ	Mono	Diverse	Δ
RIGA Cup (9-bundle)	0.70	0.061	0.214	+0.153
	0.75	0.073	0.120	+0.047
	0.80	0.055	0.050	−0.005
	0.85	0.010	0.007	−0.003
	0.90	0.001	0.000	−0.001
ACDC LV (10-bundle)	0.70	0.074	0.285	+0.211
	0.75	0.062	0.237	+0.175
	0.80	0.049	0.163	+0.114
	0.85	0.040	0.072	+0.032
	0.90	0.020	0.012	−0.007
Kvasir-SEG (11-bundle)	0.70	0.092	0.388	+0.295
	0.75	0.074	0.360	+0.285
	0.80	0.058	0.267	+0.209
	0.85	0.050	0.156	+0.106
	0.90	0.025	0.030	+0.005
BraTS foreground (2D FLAIR `seg>0`)	0.70	0.080	0.326	+0.245
	0.75	0.061	0.233	+0.172
	0.80	0.039	0.129	+0.090
	0.85	0.019	0.035	+0.016
	0.90	0.004	0.001	−0.003

Table 14: **RIGA Cup threshold diagnostic.** Values are for the released primary RIGA Cup functional pool only. Mono pools are same-bundle four-seed pools averaged over functional FMs; diverse pools are four distinct functional bundles. Percentile thresholds are computed over all model-case Dice values after the Dice$\geq$0.30 functional floor. Source: `results/_merged/diagnostics/threshold_table_riga_cup.json`.

Threshold	Mono fail $\bar{r}$	Diverse fail $\bar{r}$	Mono ALL-fail	Diverse ALL-fail
P_{25} (τ=0.513)	0.833	0.209	19.9%	2.2%
P_{33} (τ=0.575)	0.803	0.257	26.5%	5.3%
P_{50} (τ=0.686)	0.840	0.268	44.8%	15.4%

Table 15: Kvasir-SEG (polyp endoscopy) secondary 8-bundle split diagnostic — *generic-8-bundle summary*. This row uses a 1,000-image, 700/300 train/test split and is not the 11-bundle, 880/120 Kvasir row in Table 1. 7 of 8 generic encoder bundles produce functional segmentation (Dice>0.40); CLIP ViT-L/14 Dice=0.17 is non-functional and excluded from cross-family analysis. MedSAM-only traces are reported separately as an encoder-probe summary in Appendix A10; no RETFound Kvasir trace is included in the released artifact bundle.

FM Family	Dice	Within $\bar{r}$	Status
DINOv2 ViT-B/14	0.819	0.972	functional
DINOv2 ViT-S/14	0.807	0.951	functional
BiomedCLIP ViT-B/16	0.717	0.954	functional in this split
ConvNeXt-Tiny	0.651	0.949	functional
EfficientNet-B0	0.582	0.945	functional
ResNet-50	0.540	0.880	functional
ResNet-18	0.468	0.876	functional
CLIP ViT-L/14	0.174	0.839	non-functional

(rank computed across all four decoder seeds, avoiding seed-42 selection circularity; selected bundles per task are recorded in the appendix diagnostic output—e.g. Kvasir top-4 = {DINOv2-B, DINOv2-S, BiomedCLIP, ConvNeXt-T}, Cup top-4 = {BiomedCLIP, DINOv2-B, DINOv2-S, EfficientNet-B0}). This is deliberately favourable to the diverse pool and useful for a matched-size structural comparison, but it is not a deployable pool-selection rule because the ranking uses the same test set on which lift is evaluated. For each task / threshold we report three lift variants:

- **Matched-4 seed-42 (single-seed diagnostic)**: the single-seed pool; retained for comparability with a single-seed estimator.
- **Matched-4 symmetric permuted (preferred)**: each of the 4 FMs takes a *distinct* decoder seed from $\{42, 43, 44, 45\}$; the lift is averaged on the log scale over all $4! = 24$ seed-to-FM permutations, then exponentiated. This is the matched-size structural analog of the main paper's symmetric Pearson $\bar{r}$ estimator (mono = DINOv2-B with 4 distinct seeds matches matched-4 = 4 FMs with 4 distinct seeds).
- **Matched-4 symmetric independent (sensitivity)**: each FM's seed is drawn i.i.d. with replacement from $\{42..45\}$, $4^4 = 256$ tuples; reported as a robustness check (the permuted variant is preferred because mono uses four *distinct* seeds).

All bootstrap CIs are computed on *cases* only (patient-clustered on ACDC); the seed tuple space is finite and fully enumerated, not bootstrapped. We report 95% case-bootstrap CIs on the log-scale lift and exponentiate to the reported interval (B=2000). We also report the raw all-fail count ($\mathrm{AF}_{\mathrm{cnt}}$) and the Poisson-binomial independence expectation $\mathbb{E}[\mathrm{AF}_{\mathrm{ind}}]$; when $N \cdot \mathbb{E}[\mathrm{AF}_{\mathrm{ind}}] < 5$ we flag the row as support-limited. Under the matched-size comparison, the adequately supported rows are RIGA Cup τ=0.9, RIGA Disc τ=0.95, and ACDC LV τ=0.9; Kvasir τ=0.8 is mono-sparse and retained as a descriptive diagnostic to match Fig. 1. Across these selected rows the mono/matched-4 lift ratio is 1.5–$5.8\times$ (Table 16); lower-support rows give larger ratios (5.9–17$\times$) treated as descriptive order-of-magnitude only. We therefore treat τ values well below the task-specific Dice ceiling as support-limited and report them as descriptive rather than precise lift estimates.

Table 16: **Matched-M=4 lift diagnostic.** Mono is DINOv2-B $\times$ 4 seeds; matched-4 diverse uses the top-4 bundles by test-set seed-averaged mean Dice, so this is an oracle structural comparison, not a deployable selection rule. CIs are 95% case-bootstrap intervals on log-lift, exponentiated. Rows without adequate support (✓ missing) are descriptive order-of-magnitude indicators.

Task	τ	Mono [CI]	Diverse-42	Diverse-perm [CI]	Ratio*	Supp.
RIGA Cup	0.8	4450 [1408, 24012]	174 [79, 467]	**265 [130, 696]**	17$\times$	–
RIGA Cup	0.9	2.18 [1.81, 2.69]	1.33 [1.20, 1.47]	**1.44 [1.31, 1.60]**	1.5$\times$	✓
RIGA Disc	0.95	9.80 [6.82, 14.6]	2.87 [2.19, 3.82]	**3.11 [2.44, 3.98]**	3.2$\times$	✓
ACDC LV	0.5	2149 [462, 34449]	200 [51, 1026]	**188 [62, 817]**	11$\times$	–
ACDC LV	0.7	189 [55.6, 1064]	31.2 [15.5, 77.7]	**25.3 [13.1, 57.8]**	7.5$\times$	–
ACDC LV	0.8	33.8 [15.1, 107]	5.87 [3.60, 10.6]	**5.71 [3.63, 9.93]**	5.9$\times$	–
ACDC LV	0.9	2.23 [1.63, 3.32]	1.39 [1.22, 1.60]	**1.40 [1.25, 1.61]**	1.6$\times$	✓
Kvasir	0.7	246 [128, 570]	33.9 [22.3, 54.5]	**30.8 [20.4, 48.3]**	8.0$\times$	–
Kvasir	0.8	48.0 [28.9, 85.1]	8.72 [6.62, 12.0]	**8.21 [6.35, 11.1]**	5.8$\times$	mono-sparse

* Ratio = Mono / Matched-4 PERM. **Bold** = preferred symmetric-permuted estimator (24 seed-to-FM tuples). Matched-4 seed 42 is a single-seed diagnostic. Matched-4 INDEP (4^4=256 iid tuples) produces values within $\pm 1.5\%$ of PERM on every row and is omitted for compactness; the full table is a secondary diagnostic rather than the `results/_merged` primary scorer output. Supp. ✓ iff $N \cdot \mathbb{E}[\mathrm{AF}_{\mathrm{ind}}] \geq 5$ for the PERM estimator.

A10 Medical-domain encoder probe details

We implemented two medical-domain encoder paths, but only the MedSAM image-encoder probe is reported in the released evidence bundle:

RETFound is a ViT-L/16 MAE pretrained on 1.6M retinal images [Zhou et al., 2023]. The code path builds the RETFound-MAE CFP ViT-L/16 trunk and expects the gated HuggingFace checkpoint `YukunZhou/RETFound_mae_natureCFP` (or an equivalent local `RETFOUND_CFP_CHECKPOINT`); the official upstream project is `rmaphoh/RETFound_MAE`. RETFound MAE pretraining uses ImageNet $[0.485, 0.456, 0.406]/[0.229, 0.224, 0.225]$ normalisation; our frozen-probe adaptation would drop the CLS token and reshape the 196=14$\times$14 patch tokens into a spatial 14$\times$14 feature map of dimension 1024. That patch-token extraction is not RETFound's official global/CLS-style classification fine-tuning path. Because the gated checkpoint/authentication was unavailable for the reported artifact run, RETFound cells are not reported and no RETFound result supports the paper's empirical claims.

MedSAM is a SAM ViT-B fine-tuned on 1.5M medical image-mask pairs [Ma et al., 2024]. The code path uses the official `bowang-lab/MedSAM` `medsam_vit_b.pth` checkpoint through `segment_anything.sam_model_registry['vit_b']` with `MEDSAM_CHECKPOINT` pointing to that checkpoint. In this audit, we use *only the image encoder* as a frozen feature extractor with our same 1×1-conv decoder. We discard MedSAM's prompt encoder and mask decoder entirely—i.e. MedSAM enters this boundary probe on the same frozen-feature terms as the other FMs (no prompt-based path, no bounding-box input, no iterative refinement). The wrapper strips the SAM image-encoder neck and uses the pre-neck ViT-B representation: the official 1024×1024 positional grid is bicubically interpolated to the 224×224 input grid, yielding a 14×14 spatial feature map of dimension 768. This pre-neck $768\times14\times14$ tensor is an internal representation, not the official SAM/MedSAM post-neck image embedding at 1024 input. This means MedSAM in our experiment is a "MedSAM image-encoder probe", not its intended prompt-conditioned segmentation system or official inference preprocessing path. The comparison is therefore an encoder-only boundary probe for the same-bundle dependence question we ask, not a fair evaluation of MedSAM as released or of its best-case performance with its own prompt-based decoder.

All other aspects of the secondary sensitivity use the same four decoder seeds, 30-epoch Adam 10^{-3} training, and identical 1×1 conv decoder. The completed MedSAM-only probe covers ACDC LV, Kvasir, RIGA Cup, and RIGA Disc (four seeds each). Mean Dice across seeds is 0.517, 0.513, 0.734, and 0.908 respectively; the aggregate diagnostic is released as `results/_merged/diagnostics/medsam_only_summary.json`. We use this as scope context rather than primary evidence.

A11 BraTS/ACDC clustering sensitivity

The Table 1 BraTS foreground row is a BraTS 2024 GLI 2D FLAIR derivative with n=3,228 test slices from 324 held-out subjects: per held-out subject, the loader keeps up to the top-10 axial slices by foreground area, requires at least 64 positive pixels, and uses a subject-level 80/20 split with NumPy seed 42. The 3,228 count is traceable to the released `test_ids`: 321 held-out subjects contribute ten eligible slices, one contributes nine, one contributes six, and one contributes three, giving a 12-slice deficit from the 324×10 maximum. ACDC LV similarly evaluates 361 LV-positive slices nested under 20 held-out patients. The main table is therefore a case-level audit, but the released `subject_level/*.json` files include the patient/subject unit IDs linked to `test_ids` and two cluster-aware checks. **(i) Cluster bootstrap over the case-scale statistic.** Resampling patients/subjects with replacement and keeping all slices within each sampled unit gives ACDC Δ=0.321 [0.269, 0.374] and BraTS foreground Δ=0.315 [0.283, 0.351], so the case-scale gap remains positive under unit-level resampling. **(ii) Subject-aggregated statistic.** Averaging each member's per-slice Dice within patient/subject before computing Pearson gives smaller but still positive gaps: ACDC Δ=0.262 [0.151, 0.413] and BraTS foreground Δ=0.214 [0.182, 0.251]. The conservative reading is that slice dependence inflates the magnitude relative to subject means, especially on BraTS, but it does not explain the within-vs-cross separation away.

A12 BraTS sparse-class filtering caveat and released diagnostics

Per-class BraTS sensitivity can be inflated by sparse labels: if a class is absent from a slice, empty-vs-empty Dice can return a perfect score and obscure the foreground-error structure that matters for a segmentation audit. For this reason, the primary BraTS row uses the foreground derivative `seg>0` rather than a sparse NCR-only target. NCR filtering diagnostics from a separate sparse-class sweep are not used as numeric evidence here because their exact aggregate files are not included in the supplementary artifact. The released BraTS foreground train-size and multimodal diagnostics are instead reported later in Appendices A33.5–A33.6 from `results/_merged/late_brats_summaries.json`.

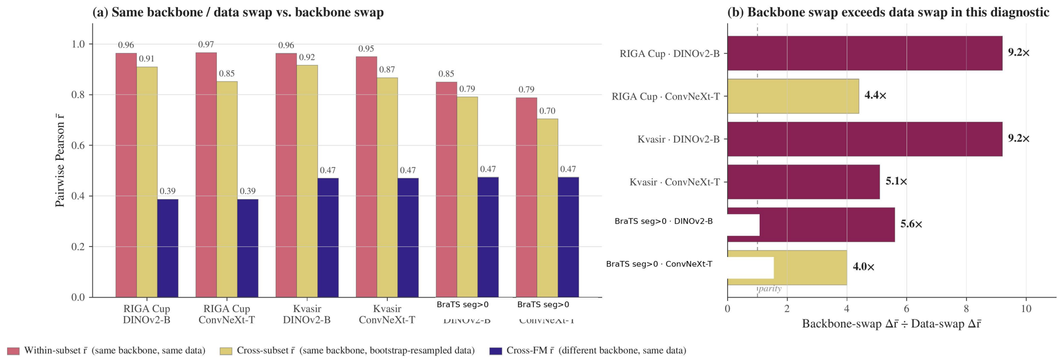

Figure 2: **Same-backbone bootstrap-bagging diagnostic.** For each shown dataset/backbone pair, we compare (i) same-backbone/same-data cross-seed correlation, (ii) same-backbone/bootstrap-resampled-data correlation, and (iii) a cross-backbone reference from the corresponding diagnostic pool. The ratio panel reports $(r_{\text{same data}} - r_{\text{cross backbone}})/(r_{\text{same data}} - r_{\text{bootstrap data}})$, so values above one mean that changing backbone reduces correlation more than bootstrap-resampling the training data in this diagnostic. The BraTS label refers to the 2D foreground derivative (`seg>0`), not the official 3D BraTS WT challenge target. This figure is descriptive and does not identify a causal mechanism.

A13 Mechanism-probe scope

This block keeps released diagnostics separate from exploratory mechanism probes whose source prediction caches are not included in the supplementary artifact. Only released traces or aggregate JSONs are used as numerical support. The shared reading is descriptive: same-backbone perturbations can reduce correlation, but the released diagnostics do not make same-backbone pools behave like the primary cross-family references.

ACDC architecture caveat. Secondary ACDC exploratory tables used a separate 8-bundle split and architecture-pair subsets that are not included as replayable aggregate artifacts. The released support for the architecture-caveat wording is instead the same-architecture-family ViT / different-pretraining diagnostic in Appendix A30 and the expanded same-family cross-checkpoint decomposition in Appendix A31.

Random-init probes. DINOv2-B random-initialisation probes used secondary splits and source prediction caches that are not included in the supplementary artifact. The released no-pretraining evidence in this bundle is limited to the CNN random-init diagnostic in Appendix A33.2, which should be read as a descriptive same-recipe sanity check rather than a causal isolation of architecture, optimiser, and data effects.

LoRA and cross-architecture probes. LoRA probes and ViT-B cross-architecture random-init sweeps require source prediction caches and aggregate summary files that are not included in the supplementary artifact. We do not tabulate their values or use them as bundle-supported evidence. The released same-backbone levers are the head-capacity sweep, UNet-skip replication, fold-reslicing diagnostic, CNN random-init diagnostic, and MC dropout.

A14 Same-backbone, different-data bootstrap-bagging audit

For each (FM, dataset) pair, four bootstrap replicates of the training set (image-level on RIGA/Kvasir; subject-level on BraTS) are crossed with four decoder seeds. Bootstrap data-swap reduces within-pool $\bar{r}$ by only 0.05–0.12, while backbone swap reduces it by 0.33–0.50. This supports the scoped claim that, in the frozen-feature protocol, resampling the training cases does not provide the same decorrelation as changing encoder family.

A15 Distribution-shift audit

We evaluate one released cross-distribution robustness diagnostic for the within–cross correlation gap using an 8-bundle RIGA pool and the frozen-feature protocol from Section 4.

RIGA leave-MESSIDOR-out (cross-vendor fundus). RIGA+ [Almazroa, 2018] combines three imaging sources: *BinRushed* (Bin Rushed Ophthalmic Center, Riyadh, Saudi Arabia), *Magrabia* (Magrabi Eye Center, Riyadh, Saudi Arabia), and *MESSIDOR* (French public corpus, different cameras and populations). We train the frozen-FM + 1×1 conv decoder on BinRushed+Magrabia (union) and evaluate on MESSIDOR (454 cases after local preprocessing/eligibility filtering from the 460-image MESSIDOR subset) as cross-vendor out-of-distribution. Per-case traces for this diagnostic are released under `results/_merged/per_case_dice/riga_cup_ood_messidor`. The released scorer uses a fixed 8-bundle $\times$ 4-seed pool without a functional-floor change between ID and OOD rows. Under that directly reproducible convention, the ID row has within/cross/gap $0.659/0.282/0.377$ and the MESSIDOR OOD row has $0.743/0.390/0.353$.

Pool-level summary. In this fixed-pool RIGA cross-vendor diagnostic, the within–cross gap changes by -0.024 and remains positive. Both within and cross correlations are higher on the MESSIDOR row, so this diagnostic should be read only as evidence that the gap survives one released cross-vendor stress test, not as evidence that distribution shift generally contracts dependence. Broader cross-scanner, cross-demographic, and longitudinal shifts remain out of scope.

Table 17: **RIGA cross-vendor diagnostic under a fixed 8-bundle scorer.** This is not case-identical to Table 1. Within = same-FM different-seed $\cup$ same-family different-FM (DINOv2-B/S and ResNet-18/50); cross = different-family different-FM. The gap remains positive, but both within and cross correlations increase on MESSIDOR; no general distribution-shift conclusion is supported. Brackets are case-level bootstrap 95% CIs (B=500). Source: `results/_merged/diagnostics/distshift_riga_messidor.json`.

Shift regime	Within-$\bar{r}$	Cross-$\bar{r}$	Gap
ID reference (fixed 8-bundle pool)	0.659 $[0.603, 0.715]$	0.282 $[0.197, 0.369]$	0.377 $[0.301, 0.463]$
OOD MESSIDOR (fixed 8-bundle pool)	0.743 $[0.722, 0.765]$	0.390 $[0.345, 0.430]$	0.353 $[0.318, 0.388]$

This diagnostic uses a paired fixed 8-bundle RIGA pool and should not be read as a case-identical comparison to the Magrabia-source held-out primary RIGA rows in Table 1.

A16 UNet-with-skip decoder replication

The paper's primary decoder is a 1×1 conv with $C{+}1$ trainable parameters for C frozen channels (for example, 769 parameters for a 768-channel feature map), which is close to a linear probe on frozen features. Dense decoders with skip connections—nnU-Net, U-Net, SegFormer—are closer to deployed medical segmentation. We therefore ran a separate diagnostic in which the frozen backbone is unchanged but the head is replaced by a multi-scale UNet-style decoder over four encoder feature levels. This is a *two-seed diagnostic grid* (seeds 42/43), not an inferential table with the same status as the four-seed primary pool; the reported correlation is the single seed-pair Pearson coefficient for each cell.

Architecture. For hierarchical CNNs (ConvNeXt-T, ResNet-50) the decoder uses the native feature pyramid as skip sources; for ViT-style encoders (DINOv2-B/S and BiomedCLIP ViT) it uses sampled intermediate token maps reassembled into a four-level pyramid. The released head first applies lateral 1×1 projections to a common decoder width, then runs a top-down bilinear-upsample path with three additive skip merges (deep $\rightarrow$ mid $\rightarrow$ shallow $\rightarrow$ shallowest), two conv-BN-ReLU blocks after each merge, a final conv-BN-ReLU after upsampling to the output grid, and a final 1×1 output conv. Training matches the frozen-feature recipe where possible: Adam lr 10^{-3}, batch 16, 30 epochs, BCE, seeds $\{42, 43\}$.

Finding. The UNet-skip head improves mean Dice substantially over several weak 1×1 probe cells, but it does not remove same-backbone co-failure: across the 20 evaluated cells, the seed-pair correlation remains 0.774–0.979. On rows where the matched 1×1 seeds 42/43 are both available, UNet-skip usually lowers seed-pair r modestly, but the effect is heterogeneous: Kvasir ResNet-50 drops $0.973\rightarrow0.774$, while RIGA Cup DINOv2-B is flat ($0.965\rightarrow0.967$) and RIGA Cup ResNet-50 increases ($0.908\rightarrow0.952$). The BraTS cells are consistent with the same boundary: DINOv2-B/S UNet-skip reaches mean Dice 0.875–0.877, but seed-pair r remains 0.886–0.936. Thus the conclusion is narrower than a decoder-family claim: a stronger skip decoder can improve accuracy and sometimes reduce seed correlation, but under this frozen-backbone two-seed grid it does not make same-backbone pools resemble the cross-family references in Table 1. A full nnU-Net v2 5-fold audit on RIGA Cup is reported in Appendix A17.

Table 18: **Two-seed UNet-skip same-backbone diagnostic.** Each cell uses the two completed seeds 42/43 from the released UNet-skip trace tree; "seed-pair r" is therefore one Pearson coefficient, not a bootstrapped four-seed within-family estimate. Primary Table 1 cross references are Kvasir 0.697, ACDC 0.639, RIGA Cup 0.361, and BraTS foreground 0.623.

Task	Backbone	Mean Dice	Seed-pair r
Kvasir	DINOv2-B	0.883	0.877
Kvasir	DINOv2-S	0.866	0.919
Kvasir	BiomedCLIP	0.806	0.926
Kvasir	ConvNeXt-T	0.860	0.775
Kvasir	ResNet-50	0.816	0.774
ACDC LV	DINOv2-B	0.864	0.979
ACDC LV	DINOv2-S	0.863	0.970
ACDC LV	BiomedCLIP	0.827	0.950
ACDC LV	ConvNeXt-T	0.885	0.917
ACDC LV	ResNet-50	0.882	0.945
RIGA Cup	DINOv2-B	0.902	0.967
RIGA Cup	DINOv2-S	0.894	0.893
RIGA Cup	BiomedCLIP	0.880	0.892
RIGA Cup	ConvNeXt-T	0.895	0.970
RIGA Cup	ResNet-50	0.881	0.952
BraTS foreground	DINOv2-B	0.877	0.936
BraTS foreground	DINOv2-S	0.875	0.886
BraTS foreground	BiomedCLIP	0.858	0.929
BraTS foreground	ConvNeXt-T	0.895	0.938
BraTS foreground	ResNet-50	0.889	0.934

A17 Fold-reslicing and nnU-Net complements

Motivation. nnU-Net [Isensee et al., 2021] is the de facto standard for medical image segmentation; its 5-fold cross-validation ensemble is a routinely-used uncertainty / robustness baseline. The full nnU-Net pipeline is *self-configuring* (2D/3D selection, preprocessing, patch-based sampling, default geometric/intensity augmentation, cascaded network, post-processing). We keep two evidence surfaces separate: (i) the released frozen-feature fold-reslicing diagnostic, which tests split sensitivity with one decoder seed per fold and therefore cannot estimate within-fold versus cross-fold same-backbone pairs; and (ii) released task-level nnU-Net summary JSONs, which are full-pipeline complements but not case-identical contrasts to the primary FM rows.

Released fold-reslicing design. This diagnostic re-slices the frozen-feature cache into 5 folds and reruns the 8-bundle pool {DINOv2-B/S, BiomedCLIP, CLIP-L, ConvNeXt-T, EfficientNet-B0, ResNet-50/18} on ACDC LV and RIGA Cup, single seed per fold cell. Because the released artifact contains one seed per fold, this diagnostic reports per-fold cross-bundle $\bar{r}$ and pool-mean Dice, not a 20-decoder fixed-backbone within/cross estimator. The released artifact provides the corresponding fold-reslicing summary and scorer.

Table 19: **Fold-reslicing cross-bundle stability, one seed/fold.** 8 bundles $\times$ 5 folds; single seed per fold cell. Cross-bundle $\bar{r}$ is the Pearson correlation averaged over the 28 bundle pairs on the held-out fold. This table cannot support within-fold versus cross-fold same-backbone claims.

Task	fold 0	fold 1	fold 2	fold 3	fold 4	mean $\pm$ std
ACDC LV	0.580	0.548	0.583	0.558	0.558	0.565 ± 0.015
RIGA Cup	0.359	0.317	0.325	0.315	0.261	0.315 ± 0.035

Finding. Train/test split sensitivity is small in this fold-reslicing check: cross-fold pool-mean Dice is stable (ACDC 0.502 ± 0.022; Cup 0.610 ± 0.012), and per-fold cross-bundle $\bar{r}$ is stable across folds (ACDC 0.565 ± 0.015; Cup 0.315 ± 0.035). This supports only the narrower statement that the cross-bundle diagnostic is not dominated by one particular 5-fold re-slicing. It does *not* support a claim about within-fold vs cross-fold same-backbone decorrelation, because those pair types require multiple decoder seeds per fold that are not present in the released traces.

Full nnU-Net v2 complement. To complement the frozen-feature proxy above and answer whether nnU-Net's full self-configuring pipeline decorrelates the pool more, we trained the full nnU-Net v2 2D U-Net on RIGA Cup with a $520/224$ split, using the generated v2 2D `nnUNetPlans` configuration after `nnUNetv2_plan_and_preprocess` (self-configured preprocessing and architecture, official default augmentation/loss schedule, 250 iterations per epoch). This split differs from the Magrabia-source held-out FM row and is therefore a task-level complement. We train two seeds per fold via custom `nnUNetTrainerSeed13_100epochs` and `nnUNetTrainerSeed37_100epochs` subclasses that pre-seed `random`, `np.random`, `torch`, and `cuda` before any nnU-Net sampler touches the RNG; each is trained for 100 nnU-Net-epochs (25,000 iterations on two A100s in parallel), then all $5 \times 2{=}10$ models predict on the held-out 224-image test set via `nnUNetv2_predict` and per-case Dice is computed against the held-out labels staged in `labelsTr`. The local nnU-Net export used a split-index file to enumerate held-out IDs; the artifact bundle releases the summary arrays/JSONs for this complement but not that local split manifest.

Across all 10 models the mean Dice is 0.949 ± 0.001 (exceeding any of our frozen-feature FM+1×1-conv combinations on RIGA Cup). The pairwise per-case Dice $\bar{r}$ is 0.972 within-fold over the 5 same-fold different-seed pairs and 0.925 cross-fold over the 40 different-fold any-seed pairs, with $\Delta_{\mathrm{within-cross}}{=}0.047$ and a case-level bootstrap 95% CI of $[0.030, 0.069]$ that excludes zero. Full nnU-Net v2 is structurally a same-architecture, different-training-data ensemble on one 2D PlainConvUNet, so it is not comparable in kind to the primary frozen-FM cross-family reference, which swaps encoder bundles. The scoped conclusion is therefore: *within this task-level same-architecture nnU-Net complement, fold identity leaves a small but positive correlation gap*; we do not claim nnU-Net substitutes for or surpasses backbone-family diversification. The 100-epoch budget, single-task coverage, and 2D-only scope are closed in Appendix A18 (1000-epoch retrain on the same RIGA Cup setup; 3D fullres nnU-Net on ACDC; higher-resolution/Dice+BCE/aug frozen-FM replication).

Data-swap comparison caveat. Appendix A14 reports bootstrap-bagging within-pool decorrelation with four decoder seeds per bootstrap replicate. The fold-reslicing row instead reports single-seed cross-bundle correlations across held-out folds. These diagnostics both stress training data/split sensitivity, but they are not dose-matched estimators and are not combined into a causal ranking.

A18 Segmentation-recipe and nnU-Net scope audit

Our main protocol uses 224×224 2D inputs, frozen backbone, 1×1-conv decoder, BCE loss, no augmentation, and 30-epoch head training. Routine medical-imaging pipelines often run with compound Dice+CE loss, geometric augmentation, 3D input, and full-length (1000-epoch) training, so we test whether the within/cross correlation structure we report is specific to the simplified protocol. We close these axes as task-level complements; the RIGA nnU-Net rows use a different 520/224 split rather than the Magrabia-source held-out FM split, so they should not be read as case-identical contrasts.

Dimensionality axis: 3D fullres nnU-Net at 1000 epochs on ACDC. Training nnU-Net v2 in its self-configuring `3d_fullres` setting on ACDC with a seed-42 $105/45$-patient split (210 training, 90 test volumes) using `nnUNetTrainerSeed{13,37}_1000epochs` (seed injected in `on_train_start` before sampler initialisation) gives 10 fully-trained 3D models across 5 folds. `nnUNetv2_predict` on all 90 held-out volumes yields per-case Dice on mean foreground and per class. Across 10 models the mean Dice is 0.920 ± 0.001, within the range reported for strong nnU-Net baselines on ACDC. On per-case mean Dice the within-fold (5 pairs) vs cross-fold (40 pairs) gap is $\Delta{=}0.043$ [0.021, 0.085] with within $\bar{r}{=}0.978$ and cross $\bar{r}{=}0.935$; per class, $\Delta_{\mathrm{RV}}{=}0.041$ [0.011, 0.110], $\Delta_{\mathrm{Myo}}{=}0.029$ [0.017, 0.046], $\Delta_{\mathrm{LV}}{=}0.067$ [0.033, 0.098]. The artifact bundle releases the aggregate summary for this 150-patient complement but not the exact split manifest; the released public ACDC converter targets patients 001–100 for the primary 2D row. We therefore treat this 3D row as a task-level scope summary rather than a from-bundle training recipe. In this ACDC nnU-Net complement, the 2D$\rightarrow$3D fullres shift changes the small fold-gap regime by less than 0.01 relative to the 2D 100-epoch baseline ($\Delta{=}0.047$ in Appendix A17).

Loss / augmentation axis: RIGA Cup frozen-FM at encoder-native 224×224. We retrain the seven-bundle RIGA Cup subset (DINOv2-B/S, CLIP ViT-L/14, ConvNeXt-T, EfficientNet-B0, ResNet-50/18) with four seeds each under a more standard segmentation loss/augmentation recipe: encoder-native input size 224×224, composite soft Dice+BCE loss, paired random horizontal flip and rotation $\pm15^\circ$, for 30 epochs. The frozen encoder wrappers use 224×224 inputs; we therefore treat this as a loss/augmentation test, not a resolution test. All other factors (frozen backbone, 1×1-conv segmentation head, Adam 10^{-3}) match the main protocol. Per-FM mean Dice ranges from 0.526 (ResNet-18) to 0.758 (DINOv2-B). Under the same symmetric estimator used for the main table, within-family $\bar{r}$ is 0.903 (58 within pairs, including same-FM seed pairs and DINOv2-B$\times$S), cross-FM $\bar{r}$ is 0.556 (320 cross pairs), and the within/cross gap is Δ=0.347 [0.275, 0.428]. The gap is smaller than the main RIGA Cup interval (0.473 [0.395, 0.556]) but remains positive; loss/augmentation changes do not remove the gap.

Training-length axis: full nnU-Net v2 at the 1000-epoch default on RIGA Cup. We retrain the 10-model RIGA Cup 2D nnU-Net pool from Appendix A17 at the nnU-Net v2 default 1000 epochs, using the same seeded trainer pattern (`nnUNetTrainerSeed{13,37}_1000epochs`) on the same $520/224$ split with identical dataset/plan files. Mean Dice across the 10 models is 0.946 ± 0.0004 (vs 0.949 ± 0.001 at 100 epochs). Within-fold $\bar{r}$=0.983 (5 pairs), cross-fold $\bar{r}$=0.949 (40 pairs), and Δ=0.034 [0.021, 0.058]. Relative to the 100-epoch run both estimators tighten slightly (within $0.972 \rightarrow 0.983$; cross $0.925 \rightarrow 0.949$) and the within/cross gap contracts from 0.047 to 0.034; measured predictions are more correlated under longer training, and the gap does not disappear. Ten-fold longer training therefore does not change the replicability structure of nnU-Net on this task-level complement; it is not a case-identical comparison to the Magrabia-source held-out FM row.

Synthesis. Across these scope analyses, the frozen-FM loss/augmentation row remains in the same large-gap regime as the RIGA Cup main row, while the same-architecture nnU-Net fold axis remains in the small-gap regime ($\Delta \approx 0.04$). Because the nnU-Net RIGA rows use a different split, the synthesis is a task-level scope check rather than a paired case-identical comparison. These rows do not support a simpler explanation that the main-table pattern is solely an artifact of BCE loss, no augmentation, short training, or the tested full-pipeline nnU-Net complements.

Table 20: **Scope audit across recipe and nnU-Net complements.** Rows are not all case-identical and should not be compared as a controlled ablation. The table asks whether selected recipe changes remove the qualitative gap. The frozen-reference row uses the same seven candidate FMs as the Dice+BCE+augmentation row before the primary functional-floor exclusion; the primary filtered RIGA Cup row is Table 1. The RIGA nnU-Net rows use a different 520/224 split and are not case-identical comparisons to the Magrabia-source held-out FM row.

Axis	Protocol	Task	Within	Cross	Δ	95% CI
Frozen reference	7-bundle candidate set, 224, BCE, no aug, 30 epochs	RIGA Cup	0.798	0.266	0.532	[0.456, 0.614]
Loss / aug	7-bundle frozen, 224, Dice+BCE, aug, 30 epochs	RIGA Cup	0.903	0.556	0.347	[0.275, 0.428]
Baseline (App. A17)	nnU-Net 2D, 100 epochs	RIGA Cup	0.972	0.925	0.047	[0.030, 0.069]
Training length	nnU-Net 2D, 1000 epochs	RIGA Cup	0.983	0.949	0.034	[0.021, 0.058]
Dimensionality	nnU-Net 3D fullres, 1000ep	ACDC (mean)	0.978	0.935	0.043	[0.021, 0.085]
	per-class RV		0.967	0.927	0.041	[0.011, 0.110]
	per-class Myo		0.987	0.957	0.029	[0.017, 0.046]
	per-class LV		0.978	0.911	0.067	[0.033, 0.098]

A19 Held-out multi-task stress test

Protocol. In addition to the four primary benchmark datasets (five task rows: Cup, Disc, ACDC, Kvasir, BraTS foreground), we evaluate five *held-out* tasks covering additional imaging domains: MSD Hippocampus [Antonelli et al., 2022] (T1 MRI, native 3-class mask {background, anterior, posterior} binarized to {background, any-hippocampus}), ISIC-2018 Task 1 Lesion Boundary [Codella et al., 2019] (dermoscopy, 2D RGB, N=80 test cases held out from a 400-image random subset of the 2,594-image training release), MSD Heart (T2 MRI cardiac), MSD Prostate (T2 MRI prostate), and MSD Spleen (CT portal). This held-out stress test uses the same 11 generic encoder bundles and four decoder seeds as the primary source pool where possible, with the same 1×1-conv decoder, no per-task hyperparameter tuning, and no model substitution. These are task-specific 2D binary derivatives, not official challenge-test evaluations. The Dice$\geq$0.30 functional floor is applied uniformly, and the released aggregate summary records 220/220 expected per-seed traces.

Interpretation. The 11-bundle held-out protocol yields three classes of outcome: (i) ISIC-2018 and Hippocampus both show direction-positive gaps under the same broad estimator family as the primary audit, but without case-bootstrap CIs and with smaller held-out support than the primary rows. (ii) Prostate and Spleen illustrate why held-out diagnostics are scope boundaries rather than

Table 21: **Held-out aggregate stress test** (11 generic encoder bundles $\times$ 4 decoder seeds; point estimates only, no case-bootstrap CIs). Hippocampus and ISIC-2018 are direction-positive, Prostate is positive but has only six test cases, Spleen is essentially near-zero despite passing the Dice floor, and Heart has no functional bundle at the Dice$\geq$0.30 floor. Tiny-N and floor-failure rows are scope boundaries and do not support the primary claim.

Task	N	Func./avail.	Within	Cross	Δ	M_{eff}
Direction-positive point estimates with enough support to read qualitatively						
MSD Hippocampus	52	11/11	0.909	0.419	**0.491**[§]	3.94
ISIC-2018[‡]	80	11/11	0.918	0.511	**0.406**[§]	2.72
Scope-limit rows: tiny held-out support or non-functional pool						
MSD Prostate	6	5/11	0.788	0.642	0.146[§]	1.23
MSD Spleen	8	11/11	0.983	0.979	0.003[§]	1.02
MSD Heart	4	0/11	—*	—*	—*	—

*No bundle passes the Dice$\geq$0.30 floor on the four-case Heart derivative, so no within/cross gap is interpretable. [§]Values are point estimates from `results/_merged/diagnostics/heldout_11fm_summary.json`; case-level CIs are not computed for this aggregate diagnostic, and the MSD Prostate/Spleen rows have too few held-out cases to support a primary claim. [‡]ISIC-2018 uses a single random N=400-image subset of the 2,594-image ISIC-2018 training release with N=80 test cases under the seed-42 80/20 split; we do not report an independent redraw and we do not use the official ISIC test server in this study.

headline evidence: Prostate is direction-positive but has only six test cases, while Spleen passes the floor yet has a near-zero gap and $M_{\text{eff}} \approx 1$ because the members fail together. (iii) Heart does not yield a functional pool under the frozen-feature 2D derivative. Thus the held-out stress test is compatible with transfer to some external rows, but it also records where the protocol is too weak or too saturated to audit same-bundle dependence.

Operational 8-bundle held-out probes. The following recovery, pool-size, abstention, and pool-selection diagnostics use an 8-bundle held-out candidate set stored under `results/_merged/diagnostics/held_out/`. They are retained as operational-method sensitivities, not as the provenance for Table 21 above.

CLIP/BiomedCLIP normalization robustness. Uniform ImageNet normalization can be a potential artifact for BiomedCLIP and CLIP-L because both have their own normalization statistics. We therefore evaluated those two FMs on Hippocampus and ISIC-2018 with the correct CLIP normalization. The released `summary_clipfix.json` reports BiomedCLIP mean Dice 0.840 (Hippocampus) / 0.908 (ISIC-2018), while CLIP-L remains below the functional floor on Hippocampus (mean Dice $<10^{-10}$) and reaches 0.106 on ISIC-2018, still below the 0.30 floor. Relative to `summary_5tasks.json`, the aggregate Δ changed by 0.002 (Hippocampus $0.731 \rightarrow 0.729$) and 0.003 (ISIC-2018 $0.552 \rightarrow 0.555$). In this held-out diagnostic, correcting CLIP/BiomedCLIP normalization does not materially change the Hippocampus/ISIC point estimates.

Held-out recovery-rate sensitivity. To complement the primary-task $\Delta_{\text{rec}} = +0.29$ (Cup) / $+0.26$ (ACDC) grand means, we computed an exploratory held-out recovery gap at τ=0.85 on the single ISIC-2018 subset: $\Delta_{\text{rec}} = +0.269$ [+0.212,+0.331] under a case-resampling calculation saved as `recovery_CI.json`. This supports the direction of the operational gap on one dermoscopy subset, but it is not a deployment-calibrated guarantee and does not resolve the held-out estimator-dependence above. Hippocampus at τ=0.85 is recovery-saturated and gives $\Delta_{\text{rec}} = -0.01$ [-0.03, $+0.00$] (diverse and mono both near-zero recovery because only 2/7 functional bundles individually pass Dice$\geq$0.85). Lower thresholds are not evaluated here.

Pool-size sweep on held-out tasks. How does diverse-vs-same-bundle conditional recovery scale with M? For $M \in \{2, 3, 4, 5, 6\}$ on the two functional held-out tasks (Table 22), diverse pools beat same-bundle seed pools at every $M \geq 3$ on the single ISIC-2018 subset (Δ_{rec}=0.27–0.46), while Hippocampus at the paper's τ=0.85 threshold is recovery-rate-saturated (all pools <0.11) because only 2 of 7 functional bundles reach Dice$\geq$0.85 individually. These held-out operational analyses are exploratory: ISIC provides one supportive dermoscopy subset, Hippocampus is threshold-saturated at τ=0.85, and the M=5, 6 same-bundle rows are proxy constructions using the available four seeds rather than true larger same-bundle pools. Released aggregate data: `results/_merged/diagnostics/held_out/pool_size_sweep.json`.

Calibration-tuned disagreement-based abstention (an exploratory alternative lever, not a conformal guarantee). A natural operational question is whether standard abstention-based uncertainty gating closes the same-bundle gap with modest additional computation. We calibrate a per-task

Table 22: **Exploratory held-out pool-size sweep.** Conditional recovery rate P[ens pass | any fail] at τ=0.85 vs pool size M. M=2 recovery is a definition artifact under the strict-majority rule, and M=5, 6 same-bundle values are four-seed proxies. This table is not a deployable pool-size recommendation.

	Hippocampus			ISIC-2018		
M	mono	diverse	Δ	mono	diverse	Δ
2[†]	0.000	0.000	+0.00	0.000	0.000	+0.00
3	0.069	0.107	+0.04	0.111	0.498	+0.39
4	0.052	0.039	−0.01	0.091	0.359	+0.27
5[‡]	0.052	0.078	+0.03	0.091	0.553	+0.46
6[‡]	0.052	0.027	−0.03	0.091	0.454	+0.36

[†] M=2 is a definition artifact under the strict-majority ($>M/2$) rule conditioned on any-fail: recovery is zero by construction whenever M=2, regardless of pool composition. We include the row for completeness. [‡] At M=5, 6 the same-bundle entries are computed on the $\min(M, S)$=4 available seeds per FM, so they do not represent true M=5, 6 same-bundle pools; the diverse entries do use legitimate M=5, 6 distinct-family compositions.

disagreement-score threshold on 20 cases (inter-member Dice standard deviation as score; grid-search the quantile q to meet a target case-risk α=0.1 on calibration; apply the selected threshold to the remaining test cases). *This is a tuned heuristic, not split-conformal prediction* [Angelopoulos and Bates, 2021]: we use a grid search over q that terminates on the first calibration risk meeting α and use inter-member variance rather than a proper conformity score, so no finite-sample risk guarantee holds. The pool is also chosen per-task on the whole task mean Dice (family-representative highest-Dice FMs), so the diverse-pool selection step leaks task labels—this biases the comparison toward diverse pools. We report it as a sensitivity. On ISIC-2018 at matched abstention $\approx$ 0.74, diverse pools reach test-risk 0.129±0.027 vs. same-bundle grand mean 0.174±0.209 (`conformal_vs_meff.json`). The *point estimate* is $\sim$ 25% lower for diverse, but the same-bundle grand-mean SD (0.21) is $\sim 8\times$ the diverse SD (0.027); the per-FM same-bundle risk means span 0.000–0.616 across the seven functional FMs, so the grand-mean comparison is highly composition-sensitive and we do not claim a tight ranking. Hippocampus cannot hit α=0.1 for either pool, giving residual risk 0.694 (diverse) vs. 0.665 (same-bundle grand mean) ($\Delta = +0.03$, with same-bundle SD 0.28, indistinguishable). A label-free conformal procedure with a proper conformity score is not evaluated here. Released aggregate data: `results/_merged/diagnostics/held_out/conformal_vs_meff.json`.

Calibration-set pool-selection probe on held-out tasks. We operationalise a small calibration exercise: given only a 20-case calibration set on a held-out task, can a rule pick a 4-of-8 FM pool from the same secondary held-out candidate set that beats random? We evaluate five rules—(R1) random, (R2) top-4 by mean Dice on calibration, (R3) max-M_{eff}, (R4) quality-gated max-M_{eff} (Dice$\geq$0.30 floor then max-M_{eff} among qualifying), (R5) diverse-by-architecture (2 best-Dice ViT + 2 best-Dice CNN; reported as an exploratory hand-crafted baseline, not a general proposal)—and report conditional recovery rate P[ens pass | any fail] at τ=0.85 on the remaining held-out cases, averaged over 100 calibration-set bootstraps (Table 23). All rule selectors and the recovery estimator use seed-42 per-case Dice (not seed-averaged), matching a one-model-per-FM pool construction. Hippocampus has lower effective recovery support than ISIC at τ=0.85, so its recovery numbers are exploratory. In this diagnostic, simple top-4-by-Dice has the highest mean on both tasks (Hippocampus 0.161 vs. 0.043 random; ISIC-2018 0.581 vs. 0.378 random), while max-M_{eff} alone is worse than random (0.011 / 0.327). This supports the narrower point that M_{eff} is a dependence readout rather than a standalone operational objective; it is not a validated clinical pool-selection rule. Released aggregate data: `results/_merged/diagnostics/held_out/pool_selection_challenge.json`.

Held-out estimator note. Split-stability and alternate-estimator numerical summaries whose exact aggregate files are not included in the supplementary artifact are omitted from the numeric evidence. This section therefore relies on the released 11-bundle held-out aggregate and the released operational JSONs listed above.

ISIC-2018 single-subset note. The ISIC-2018 row above is a single random N=400-image subset of the ISIC-2018 Task 1 training release with N=80 held-out test cases under a seed-42 80/20 split. We do not report an independent replication of this subset; a fully random redraw of the subset or use of the official ISIC evaluation server would be a natural next step but is out of scope for this paper.

Table 23: **Exploratory pool-selection probe.** Conditional recovery rate at τ=0.85 (mean $\pm$ SD over 100 calibration bootstraps, 20 cases each, using seed-42 traces). Selection uses calibration labels. This probes whether M_{eff} alone is a poor objective; it is not a validated clinical selection rule.

Rule	Hippocampus	ISIC-2018
(R1) Random 4-of-8	0.043 ± 0.050	0.378 ± 0.125
(R2) Top-4 by calibration Dice	**0.161** ± 0.017	**0.581** ± 0.030
(R3) Max M_{eff}	0.011 ± 0.011	0.327 ± 0.100
(R4) Quality-gated max M_{eff}	0.016 ± 0.014	0.381 ± 0.112
(R5) Diverse-by-arch (2 ViT + 2 CNN)	0.096 ± 0.017	0.445 ± 0.036

A20 MC dropout diagnostic scope

A separate MC-dropout probe suggested lower any-draw correlations, but its traces are not included in the supplementary artifact and its scorer differs from the released averaged-readout diagnostic. We therefore do not tabulate it or use it as evidence. The released MC-dropout diagnostic is Appendix A33.7: under the deterministic-vs-MC-averaged readout, p=0.1–0.4 reduces within-FM $\bar{r}$ only from roughly 0.97–0.99 to 0.95–0.89, so MC dropout is a weak within-family lever in this diagnostic and does not approach the tested backbone-family swap.

A21 Full fine-tuning scope

RIGA Cup full fine-tuning epoch-sweep traces and aggregate summaries are not included in the supplementary artifact. We therefore do not tabulate those values or use them as numerical evidence. The released adaptation evidence is limited to released traces and summaries named elsewhere in the appendix, including a small set of ACDC DINOv2-B full-fine-tuning provenance traces but not the RIGA epoch-sweep matrix; full-fine-tuning numbers not backed by released paths should be read only as exploratory mechanism notes and not as support for the paper's claims.

A22 Same-backbone and scope diagnostic index

Table 24: **Qualitative index of same-backbone and scope diagnostics**. Rows are heterogeneous and not CI-comparable; the table is a navigation aid, not an ordered causal hierarchy. The RIGA-oriented map is anchored on the DINOv2-B 4-seed baseline ($\bar{r}\approx$0.96) unless marked. LoRA, full-fine-tuning epoch-sweep, and ViT random-init probes whose traces are not distributed are omitted.

Diagnostic	Readout	Approx. change from stated baseline	Evidence
Mono baseline (same FM, 4 seeds)	0.96	0.00	RIGA diagnostic baseline
UNet-skip decoder (5 backbones, 4 tasks)	0.774–0.979	heterogeneous	Appx. A16, Tab. 18
Fold-reslicing cross-bundle check	0.315–0.565	split-stable	Appx. A17; released summary
MC dropout p=0.1–0.4	0.89–0.95	-0.02 to -0.10	Appx. A33.7
Same-backbone bootstrap-bagging	0.85–0.92	-0.05 to -0.12	Fig. 2
RIGA cross-vendor shift	OOD w/c 0.743/0.390	gap 0.353; $\Delta_{\text{gap}}-0.024$	Appx. A15
Decoder capacity sweep	$\sim$ 0.69	~ -0.27	Appx. A33.1
Cross-family pretrained 8-bundle pool	**0.38** [0.29, 0.46]	**−0.58**	RIGA mechanism diagnostic

A23 Leave-one-family-out scope

Leave-one-family-out summaries were used as exploratory robustness checks, but the exact aggregate source is not included in the supplementary artifact. We therefore omit the numeric table from the released evidence record. The primary robustness check for the primary rows is the released `diagnostics/audit_sensitivity.json`, which includes leave-one-task and leave-one-FM summaries under the same scorer path.

A24 Joint-failure lift at support limits

Joint-failure lift $L{=}P_{\text{empirical joint}}/\prod_m P_{\text{marginal}}$ is unstable when marginal fail rates are near 0 or near 1. When $P_{\text{marginal}} \approx 0.3$, independence-implied joint ≈ 0.01; small sampling variation in the empirical joint produces large lift ratios. Representative near-ceiling cells: Only the ACDC $2.8\times$ row is adequately supported by the $N \cdot \mathbb{E}[\mathrm{AF}_{\mathrm{ind}}] \geq 5$ rule. The RIGA Cup $6.1\times$ and $38.8\times$ rows are support-limited, and the $123.2\times$ and $189\times$ rows are extreme sparse-tail diagnostics.

Table 25: **Support-limit examples for joint-failure lift.** Extreme lifts are shown to explain instability, not to support a headline claim. Lifts $\geq 50\times$ are descriptive (support-limited, wide noise envelope) and are *not* paper-headline statistics.

Task / Pool	τ	Indep. joint	Empirical joint	Lift
RIGA Cup mono BiomedCLIP	0.85	0.0010	0.125	$123.2\times$
RIGA Cup mono DINOv2-B	0.85	0.0047	0.183	$38.8\times$
RIGA Cup diverse 4-bundle	0.85	0.0160	0.098	$6.1\times$
ACDC LV mono DINOv2-B	0.70	0.0002	0.038	$189\times$
ACDC LV diverse 4-bundle	0.85	0.135	0.378	$2.8\times$

A25 3D segmentation-architecture pilot

This section provides scope context only. Its per-case outputs and scorer version are not included in the supplementary artifact, so the pilot is not used as primary evidence for Table 1, the abstract, or the released artifact claims. It motivates why we avoid generalizing the frozen 2D FM-pool result to 3D segmentation-architecture pools.

We assembled a 3D segmentation-architecture pool for BraTS 2024 3D volumes (271 subjects, 216/55 train/test, seed 42) using three architectures in the implementation path: SegResNet (optionally initialised from the MONAI BraTS bundle), DynUNet (random-init), and UNETR (random-init). SwinUNETR random-init pilot cells were included initially but dropped after degenerate predictions across all seeds. For each retained (architecture, seed $\in \{42, 43, 44, 45\}$) cell, the pilot trained the full MONAI segmentation model for 100 epochs with Adam 10^{-3} and BCE loss; UNETR-style models used sliding-window inference where required by memory. Reported at three filter modes:

Table 26: **3D pilot for scope context only.** Per-case outputs and scorer version are not included in the supplementary artifact; this table is excluded from the abstract and main-result evidence. Three functional-filter modes are applied uniformly in the pilot. Strict filter (Dice ≥ 0.50) retains only SegResNet pretrained and DynUNet random-init and yields $\Delta{=}0.045$, substantially below the 2D BraTS foreground $\Delta{=}0.315$.

Filter	FMs retained	Within	Cross	Δ [95% CI]
Mixed (all cells)	3 (12 cells)	0.772	0.595	0.177 [0.117, 0.248]
Dice ≥ 0.30	3 (12 cells, drop UNETR seed 43 degenerate)	0.919	0.703	0.216 [0.134, 0.271]
Dice ≥ 0.50 (strict functional)	2 (SegResNet pretrained, DynUNet random)	0.916	0.871	**0.045** [0.015, 0.112]
Random-init only (DynUNet + UNETR)	2 random-init	0.724	0.479	0.245 [0.155, 0.351]

Interpretation. Within the strict functional regime (only converged pretrained + converged random-init CNN-style 3D architectures), the 3D segmentation-architecture pilot shows $\Delta{=}0.045$—substantially below the 2D BraTS foreground $\Delta{=}0.315$. We treat this only as a scope warning: the 2D frozen-feature same-bundle dependence claim should not be extended to 3D segmentation-architecture pools without a released, case-identical 3D benchmark. Random-init 3D architectures cross-correlate *more* than pretrained 3D architectures (within random 0.724 vs cross-architecture random 0.479, $\Delta{=}0.245$), so architecture-family effects again appear large in this pilot. The degenerate SwinUNETR pilot cells limit architecture coverage in the 3D pilot; expanded 3D foundation-model evaluation (Models Genesis, Vista3D, Triad-MRI) is not evaluated here.

A26 Matched-recipe adaptation scope

Matched-recipe adaptation factorials require source prediction caches and aggregate summary JSONs that are not included in the supplementary artifact. We therefore do not tabulate those values and do not use them as artifact-supported evidence. The released adaptation evidence is limited to explicit released traces under `per_case_dice_adapted` and the aggregate summaries named in the surrounding appendix sections.

A27 Item-difficulty residualisation and held-out nulls

The main-text Section 5 reports a cross-fitted residualised Δ diagnostic on all five primary rows and keeps the two-task held-out-partition diagnostic below as a leakage warning.

Naive (leaked) null. Define difficulty $d_i = 1-\frac{1}{K}\sum_{k=1}^{K} D_k^{(42)}(i)$ where $D_k^{(42)}$ is bundle k's per-case Dice at seed 42, then residualise each $D_k^{(s)}$ against d by OLS and then evaluate $\bar{r}$. This instantiates one item-difficulty null inspired by Jo et al. [2026], but has a structural flaw: each bundle's Dice is a component of its own regressor (since d averages over all 8 bundles including k), so residuals are mechanically constrained to sum to approximately zero across bundles at each case. For K bundles at seed 42, cross-bundle residual correlations are pushed toward $-1/(K-1) \approx -0.14$, and we observe cross-family residuals of -0.088 (Cup) and -0.104 (ACDC), which sharply widens Δ compared to the raw statistic. This is a mechanical leakage artifact, not a cleaner measurement.

Two-task held-out-partition null. To remove the leakage we split the 8 bundles into halves A and B, estimate d from A's seed-42 Dice only, and residualise B's Dice on this d. The $\bar{r}$ statistics and Δ are then computed on B. The released diagnostic JSON records a fixed set of 10 held-out partitions and reports the mean and range of the resulting family-grouped Δ values across those partitions.

Table 27: **Two-task item-difficulty residualisation diagnostics.** The naive null is included only as a leakage warning. Raw values use this diagnostic's family grouping and are not Table 1 values; the held-out-partition null avoids leakage on Cup/ACDC only. Source: `r7_item_difficulty_holdout.json`.

Null	RIGA Cup Δ	ACDC Δ	Notes
Raw	0.499	0.600	Family grouping
Naive difficulty (all-bundle leaked)	0.886	0.772	Cross-family residuals$\approx$ -0.10 (leakage signature)
Held-out difficulty (released held-out partitions)	**0.639**	**0.622**	Family grouping; n=10 per task
Held-out range	[0.482, 0.823]	[0.472, 0.941]	Min / max across partitions

The two-task held-out estimate is above raw Δ on Cup ($+0.14$) and remains slightly above raw on ACDC ($+0.02$). Under this *one-directional* null, item difficulty does not explain the gap away. The naive inflation was a mechanical leakage artifact regardless.

All-row cross-fitted diagnostic. Because the held-out-partition diagnostic above covers only Cup and ACDC, the released `item_difficulty_robustness.json` adds an all-row cross-fitted diagnostic. For each trace, item difficulty is estimated from traces outside that trace's broad upstream family, that trace is residualised by OLS, and the Table 1 same/cross rule is recomputed on residuals. The residualised gap is positive on all five primary rows (Table 28). Residualisation changes the scale of correlations and can increase gaps, so this is a sign and attenuation/scale-sensitivity check, not a causal decomposition.

Table 28: **Cross-fitted item-difficulty residualisation.** Residualisation changes the correlation scale and can increase gaps; values are used only to test whether held-out item difficulty absorbs the same/cross separation. Each row uses the paper's Dice$\geq$0.30 functional pool and B=2,000 case bootstrap intervals from `item_difficulty_robustness.json`.

Task	Raw Δ	Res. same	Res. cross	Res. Δ	Res. CI	Mean R^2
RIGA Disc	0.638	0.854	-0.020	0.874	[0.805, 0.920]	0.138
RIGA Cup	0.473	0.750	-0.081	0.830	[0.782, 0.882]	0.318
Kvasir polyp	0.261	0.871	-0.089	0.960	[0.895, 1.008]	0.663
ACDC LV	0.321	0.880	-0.096	0.976	[0.950, 1.000]	0.607
BraTS foreground	0.315	0.843	-0.079	0.921	[0.910, 0.932]	0.574

Residualisation caveat. The released held-out partitions are one-directional diagnostics rather than a symmetric enumeration over every possible difficulty/evaluation split. They should be read as evidence that this particular leakage-free residualisation does not absorb the gap, not as a two-sided causal decomposition of difficulty, architecture, and pretraining. The all-row cross-fitted diagnostic broadens coverage to every primary row, but it still estimates difficulty from the same benchmark panel and therefore remains a robustness check rather than a pre-registered causal adjustment.

A28 Correlation-equivalent effective pool size M_{eff}

Definition. For a pool of M members whose per-case failure indicators $F_1, \ldots, F_M$ at threshold τ are exchangeable with mean pairwise Pearson correlation ρ_{fail}, the variance of the per-case mean indicator is $\sigma_F^2[1 + (M-1)\rho_{\text{fail}}]/M$ where σ_F^2 is the marginal variance. Equating this with the variance of a hypothetical independent pool of size M_{eff}, i.e. $\sigma_F^2/M_{\text{eff}}$, gives

$$M_{\text{eff}} \;=\; \frac{M}{1 + (M-1)\,\rho_{\text{fail}}}.$$

This is the equicorrelation approximation also used in climate-model ensembles ("effective number of models"). We report it as a second-moment readout of the audit; it is *not* a literal independent-member count.

Table 29: **Threshold-specific effective pool-size diagnostic at τ=0.85.** M_{eff} uses an equicorrelation approximation for binary failure indicators and is not a literal independent-member count. Same-bundle pools are each FM's completed decoder seeds. Diverse rows enumerate every 4-bundle composition of the filtered functional pool and every available one-seed-per-FM tuple. Undefined zero-variance failure-correlation tuples are excluded rather than imputed. Source: `results/_merged/m_eff/*.json`.

Task	Func. bundles	Mono mean†	Diverse ρ_{fail}	Diverse M_{eff}	Valid tuples
Kvasir	11	1.12	0.333±0.104	2.05±0.32	84,480 / 84,480
ACDC LV	10	1.14	0.298±0.075	2.14±0.24	53,760 / 53,760
BraTS foreground	10	1.13	0.273±0.089	2.24±0.32	53,760 / 53,760
RIGA Cup	9	1.16	0.246±0.121	2.43±0.63	31,488 / 32,256
RIGA Disc	11	1.17	0.126±0.067	2.97±0.44	84,480 / 84,480

†Mean across per-FM seed pools after excluding undefined zero-variance failure indicators. The RIGA Cup row has 7/9 defined mono pools and the RIGA Disc row has 10/11; the other rows have all mono pools defined.

Reading the table. The scorer uses the same estimand as the text: a diverse pool is four distinct FMs with one decoder-seed member from each FM. Under that definition, the nominal M=4 diverse pools yield $M_{\text{eff}} \approx$ 2–3 under this equicorrelation readout, not four; same-FM seed pools are near one under the same calculation. The exact level varies by task and failure threshold, so we use M_{eff} as a dependence readout, not as a literal deployment objective.

Pairwise ρ_{fail} heterogeneity disclosure (equicorrelation caveat). The M_{eff} formula assumes constant pairwise correlation across member pairs (equicorrelated pool). In practice the per-pair ρ_{fail} varies across the bundles in each task. From the `m_eff/*.json` outputs, mono-pool ρ_{fail} across the per-task functional bundles has mean / SD / range: Kvasir mean 0.865, SD 0.103, range $[0.601, 0.948]$ (n=11); ACDC LV mean 0.847, SD 0.122, range $[0.518, 0.931]$ (n=10); BraTS foreground mean 0.860, SD 0.117, range $[0.553, 0.958]$ (n=10); RIGA Cup mean 0.833 over 7 defined mono pools, and RIGA Disc mean 0.812 over 10 defined mono pools. Undefined rows occur when an FM's τ=0.85 failure indicator is constant, giving zero-variance Pearson. Diverse 4-of-K compositions span a wider ρ_{fail} distribution, which is why Table 29 reports means and standard deviations across all valid seed tuples rather than selecting a representative pool.

A29 Shared-backbone random-effects representation

Model. For each FM m in a frozen-feature pool and test case i, let the per-case score (or its logit) decompose as

$$s_m(i) \;=\; u(i) \;+\; a_{\text{arch}(m)}(i) \;+\; p_{\text{family}(m)}(i) \;+\; \varepsilon_m(i),$$

where u is task-inherent item difficulty, a is a zero-mean architecture-family random effect, p is a zero-mean pretraining-family random effect nested within architecture, and ε_m is an FM-specific adaptation noise. Assume u, a, p, ε are mutually uncorrelated with variances $\sigma_u^2, \sigma_a^2, \sigma_p^2, \sigma_\varepsilon^2$. Then the pairwise covariance of $s_m, s_{m'}$ takes a nested form:

$$\mathrm{Cov}(s_m, s_{m'}) \;=\; \begin{cases} \sigma_u^2 + \sigma_a^2 + \sigma_p^2, & \text{(same family)} \\ \sigma_u^2 + \sigma_a^2, & \text{(same architecture, different pretraining)} \\ \sigma_u^2, & \text{(different architecture).} \end{cases}$$

Diagnostic expectation. Under this second-moment description, and absent parameter cancellations, one expects the ordering

$$\rho_{\text{within-FM}} \;\geq\; \rho_{\text{same-arch-diff-pretraining}} \;\geq\; \rho_{\text{cross-arch}}$$

as a qualitative diagnostic rather than a theorem. Empirically on 5 same-architecture-family ViT variants (Table 3) we observe the ordering on Cup, and on ACDC after applying the functional filter: $0.988 \geq 0.807 \geq 0.639$; the all-5 ACDC row is a sensitivity to including nonfunctional CLIP-B. This is consistent with the predicted ordering and with trajectory/same-basin evidence from deep ensembles and model soups [Fort et al., 2019, Neyshabur et al., 2020, Wortsman et al., 2022]; Abe et al. [2024] is better read here as motivation that predictive diversity can remain constrained in high-capacity ensembles, not as a trajectory-bound theorem.

Implication for M_{eff}. When the binary failure thresholds preserve the same broad ordering as the continuous Dice scores, M_{eff} is monotone decreasing in the observed ρ_{fail} by definition. The empirical diagnostics therefore suggest the descriptive ordering $M_{\text{eff}}^{\text{within-FM}} \leq M_{\text{eff}}^{\text{same-arch}} \leq M_{\text{eff}}^{\text{cross-arch}}$ in the tested regime, but the paper estimates this quantity from released per-case traces rather than inferring it from labels alone. Same-bundle seed pools are same-FM by construction and sit near the lower end of the measured M_{eff} range; diverse pools that cross both architecture and pretraining bundles sit closer to the upper end.

Caveats. This is a second-moment model at the per-case-score level, not a first-principles claim about SGD dynamics. The random-effects representation is descriptive; it gives a language for the observed ordering and a hook for bounding pool redundancy, not a proof that all shared-pipeline pools must have the same structure. The item-difficulty residualisation of Appendix A27 removes u and leaves $a + p + \varepsilon$; our held-out null is consistent with $a + p$ contributing after u is removed.

A30 Architecture-confound diagnostics

The main-text Table 3 reports the released 5-variant same-architecture-family ViT / different-pretraining diagnostic on Cup and ACDC. A secondary Kvasir-SEG replication used the same 5 ViT-B variants, but its aggregate JSON is not distributed with the supplementary artifact, so it is not tabulated as released evidence here.

Table 30: **Appendix detail for the same-architecture ViT diagnostic.** Within-FM is the decoder-seed floor; same-architecture-family ViT cross-pretraining holds the broad architecture family fixed while varying pretraining. This expands Table 3 by showing the all-5 ACDC sensitivity and the functional-filtered readout. Source: released aggregate diagnostic `results/_merged/diagnostics/r7_arch_confound_analysis.json`; raw NPZ prediction caches are not included.

Task	ViT subset	Used	Seed floor	ViT cross-pretrain	Primary cross	Gap
RIGA Cup	all 5 ViTs	5/5	0.968	0.680	0.361	0.288
ACDC LV	all 5 ViTs	5/5	0.965	0.631	0.639	0.334
ACDC LV †	CLIP-B dropped	4/5	0.988	0.807	0.639	**0.182**

The same-architecture cross-pretraining separation is 0.18–0.29 across the released Cup/ACDC diagnostics under the paper's functional filter († ACDC row drops the non-functional CLIP-B at Dice=0.21<0.30, which otherwise compresses same-arch cross $\bar{r}$ and inflates the separation to 0.33). The direction is preserved, but the claim is mechanism-diagnostic rather than causal. Results are consistent with the random-effects representation in Appendix A29.

A31 Same-family cross-checkpoint decomposition

The main text explicitly cautions that the primary "same" column is dominated by same-bundle different-decoder-seed pairs. To test whether the signal collapses to that seed floor, we recompute the released same-task per-case traces, including additional same-family checkpoint variants beyond the primary pool, under a three-stratum decomposition: (i) same-FM decoder-seed pairs, (ii) distinct checkpoints within a broad upstream family (DINOv2-B/L/S, OpenAI CLIP B/16–B/32–L/14, ConvNeXt-T/S/B, and ResNet-18/34/50/101 where functional), and (iii) cross-family pairs. This is a diagnostic broad-family grouping rather than a causal architecture/pretraining decomposition. The distinct-checkpoint same-family gap remains positive on all five primary rows, but it is substantially below the same-FM seed floor.

Table 31: **Same-family cross-checkpoint diagnostic.** Same-FM seed pairs form the upper dependence floor; distinct-checkpoint same-family pairs remain above the diagnostic cross-family stratum but below same-FM seeds. The `Diagnostic cross` column is not the Table 1 cross column. Source: `results/_merged/diagnostics/cross_checkpoint_family.json`.

Task	Same-FM	Xckpt same-family	Diagnostic cross	Xckpt–cross	95% CI
Kvasir	0.987	0.777	0.697	0.080	[0.043, 0.122]
ACDC LV	0.986	0.750	0.616	0.134	[0.113, 0.155]
BraTS foreground	0.977	0.682	0.589	0.093	[0.085, 0.100]
RIGA Cup	0.963	0.537	0.271	0.267	[0.204, 0.332]
RIGA Disc	0.971	0.421	0.194	0.227	[0.149, 0.293]

CIs are case-bootstrap intervals with B=2,000. The cross column is the diagnostic broad-family cross stratum from this expanded checkpoint grid, not the Table 1 analytic-pool cross-family column. For the slice-derived rows, subject-aggregated Xckpt–cross gaps remain positive: ACDC 0.166 [0.091, 0.262] and BraTS 0.098 [0.078, 0.119]. BiomedCLIP is kept separate from OpenAI CLIP because it changes both pretraining corpus and objective. The OpenAI CLIP same-family row mixes CLIP-B/16 QuickGELU-style checkpoints with the released CLIP-L/14 `open_clip` standard-GELU dense wrapper, so that family-specific xckpt readout also absorbs this loader/activation difference.

A32 Release manifest and artifact metadata

The artifact bundle includes the primary-task result JSONs, source code, metadata, selected figure / appendix inputs, and selected aggregate diagnostic JSONs needed to audit the numerical claims from saved artifacts where redistribution is permitted. The primary rows are stored under `results/_merged`: the per-seed `per_case_dice` JSONs carry `test_ids`, `per_case_dice`, FM name, seed, and checksum fields; `within_cross/*.json`, `m_eff/*.json`, and `paper_table.json` store the primary summaries. The same directory also includes `subject_level/*.json` for the ACDC/BraTS cluster checks, `calibration_split/*.json` as split-half sensitivity (not the primary protocol), `provenance_summary.json` for source counts and case-unit metadata, and `diagnostics/*.json` for the pixel-error, matched-lift, architecture, quality-controlled pair-regression, item-difficulty, and held-out aggregate diagnostics used only as secondary evidence. The bundle does *not* include raw medical images, raw labels, full prediction NPZs, feature caches, or held-out raw prediction arrays; full split manifests beyond the case IDs embedded in the result JSONs are not included in this artifact release.

ID and source-label scope. Kvasir rows expose only image-level file identifiers in the released `test_ids`; no patient, video, or near-duplicate grouping identifier is available in the released trace bundle, so we do not claim patient-level duplicate control for that row. RIGA Cup/Disc rows retain alignment identifiers and source labels only to make the n=95 Magrabia-source test split auditable. Some upstream public/research identifiers may inherit folder-name tokens from the source release; these tokens are not validated demographic annotations and must not be used for subgroup, gender, or individual-level analysis.

Encoder-loader boundary. The released traces should be read as frozen-feature probes through the bundled loader wrappers, not as official vendor inference runs. In particular, the `clip_vitl14` traces use the version-pinned `ViT-L-14/openai` path exposed by the installed `open_clip` package; `open_clip` also exposes a `ViT-L-14-quickgelu/openai` model definition and warns that OpenAI weights are associated with QuickGELU. We therefore treat CLIP-L/14 as the bundle's documented dense-feature wrapper and do not use it to make a claim about official OpenAI CLIP inference accuracy. The code keeps this loader unchanged so that the released per-case traces remain reproducible.

Compute and environment scope. The artifact-verification path is CPU-only: the bundled verification script and the individual `compute_*` scripts read released JSON traces and finish in minutes on a typical workstation. Raw-image reproduction is heavier and requires separate upstream data/checkpoint access. The reported training runs used a shared GPU cluster with NVIDIA A100-80GB, A6000, and L40S devices; feature extraction time depends on backbone and task size, and cached frozen-head training is lightweight relative to feature extraction. We therefore document resource classes and released trace-level commands rather than claiming a single exact GPU-hour total because raw-image retraining is outside the submitted artifact's reproduction scope.

Table 32: **Compute scope for reproducibility.** The submission supports trace-level recomputation directly; raw-image training resources are provided as reproduction guidance because upstream data, checkpoints, and feature caches are not redistributed.

Reproduction target	Required compute	Notes
Primary and diagnostic summary JSONs	CPU, minutes-scale	Run `PYTHONPATH=code/src python3 code/scripts/verify_artifact.py`; no raw images, checkpoints, Torch, or GPU required.
Frozen-feature primary traces from raw data	A100-80GB/A6000/L40S-class GPU plus local storage for upstream data and features	Feature extraction is the main cost; cached 1×1 decoder-head training uses Adam for 30 epochs and is comparatively lightweight.
Appendix architecture, adaptation, and nnU-Net complements	Same cluster GPU classes; optional MONAI/nnU-Net environments where applicable	These are diagnostic or limited-scope runs. The bundled trace-level artifact is sufficient to recompute reported released summaries without rerunning them.

Table 33: **Trace-level claim-to-artifact map.** The artifact supports recomputation of primary audit statistics from derived per-case Dice traces. It does not reproduce raw-image training unless users separately obtain upstream datasets and checkpoints. Aggregate-only diagnostics are marked as such rather than promoted to primary evidence; exact commands and checksums are in `README.md`, `metadata/MANIFEST.md`, and `metadata/SHA256SUMS`.

Claim/readout	Released source	Recompute / limitation
Primary same-vs-cross gaps, CIs, and filtered functional pools	`paper_table.json`, `within_cross/*.json`, `per_case_dice/*.json`	`PYTHONPATH=code/src python3 code/scripts/verify_artifact.py`; case IDs and per-case Dice are included, raw images and masks are not; trace-level recomputation only.
Subject, floor, ICC, permutation, event, and quality-control robustness	`subject_level/*.json`, `audit_sensitivity.json`, `failure_event_summary.json`, `quality_controlled_pairs.json`	The verification script calls `compute_all.py`, `compute_audit_sensitivity.py`, `compute_failure_event_summary.py`, `compute_quality_controlled_pairs.py`; these are estimator checks, not new training runs.
Item-difficulty and cross-checkpoint diagnostics	`item_difficulty_robustness.json`, `cross_checkpoint_family.json`, `r7_arch_confound_analysis.json`	`compute_item_difficulty_robustness.py` and `compute_cross_checkpoint_family.py`; diagnostic evidence for alternative explanations, not causal identification.
Pixel/lift, same-backbone, held-out, and nnU-Net scope rows	`diagnostics/pixel_error/*.json`, `diagnostics/matched_lift/*.json`, appendix trace files, `nnunet/*.json`	Aggregate or task-level support only where raw masks/prediction caches or case-identical full-pipeline comparators are not included.

Table 34: **Trace inventory and analytic-pool provenance.** `Merged` counts all completed (bundle, decoder-seed) JSON entries under `results/_merged/per_case_dice`; Table 1 uses the original 44-trace primary source subset and applies the functional-floor pool shown in the final column. Same/cross pair counts for the analytic pool are reported in Table 1. The release key `brats_wt` denotes the 2D BraTS `seg>0` foreground derivative used in this paper; it is not the official 3D BraTS WT challenge target.

Task	*N*	Merged bundles	Merged members	Table 1 source	Analytic pool
`kvasir`	120	17	68	44	11 bundles
`acdc_lv`	361	17	68	44	10 bundles
`riga_cup`	95	17	68	44	9 bundles
`riga_disc`	95	17	68	44	11 bundles
`brats_wt`	3228	17	68	44	10 bundles / 40 members

Functional-floor members are retained in the source files. The paper applies a Dice$\geq$0.30 functional floor downstream in the estimator, not at result ingestion. Thus the merged `per_case_dice` tree can list more completed JSONs than the analytic pool used by Table 1; the extra cross-checkpoint members support Appendix A31. For example, ACDC LV has 68 merged source JSONs but Table 1 retains the 44-trace primary source subset and 10 bundles after the floor. The authoritative row-level provenance for the table is `within_cross/*.json`, especially the `pool` and `fms_kept_after_floor` fields.

What else is in the bundle. In addition to the primary `per_case_dice` tree, the bundle includes released trace trees for decoder capacity, UNet-skip decoding, loss/augmentation sensitivity, fold-reslicing, BraTS train-size, MC dropout, BraTS multimodal input, random-init adapted heads, and a small set of ACDC DINOv2-B full-fine-tuning provenance traces. It also includes selected aggregate diagnostic JSONs under `results/_merged/diagnostics` plus code sufficient to inspect and regenerate the included primary summaries and appendix summaries. Appendix-only diagnostics not listed here, raw prediction arrays, and full pixel-mask caches are not included in the supplement and are treated as reported diagnostics rather than fully bundled reproduction targets.

What is not in the bundle and why. Raw medical images, raw labels, feature caches, and full prediction NPZ caches are not included. For restricted datasets such as BraTS, the release is limited to derived per-case Dice and aggregate dependence readouts; users must obtain the raw data from the upstream provider. Larger derived caches and fuller split manifests are not included in the released artifact.

A33 Additional mechanism and scope diagnostics

This section reports seven additional scope analyses that close axes around the main result: decoder capacity, random-initialized CNNs, loss/augmentation, fold-reslicing, BraTS train-size, BraTS multimodal input, and MC dropout. The released artifact contains the corresponding per-case Dice traces and the CPU-only summary scripts used for these appendix tables. All numbers are computed directly from the per-case traces; per-FM means use seed averaging where multiple seeds are present, and Pearson $\bar{r}$ is symmetric multi-seed.

A33.1 Decoder-capacity sweep

We rerun the Section 6 (C1) decoder sweep at four explicit head designs spanning 769 to 3.94M trainable parameters: `linear_1x1`, `mlp_2layer`, `unet_lite`, and `transformer_decoder`. Two FMs (DINOv2-B, BiomedCLIP) $\times$ two seeds $\{42, 43\}$ on RIGA Cup. Across the $\sim 5{,}000\times$ capacity range, pool-mean Dice ranges over 0.725–0.848 but cross-decoder within-FM Pearson $\bar{r}$ remains 0.69 on average (n=48 pairs across the four head designs and the two FMs)—above the RIGA Cup functional-pool cross-family reference of 0.361 (Table 1). Head capacity changes accuracy and some correlations, but does not eliminate same-backbone co-failure in this diagnostic.

Table 35: **Head-capacity sweep on RIGA Cup** (2 FMs $\times$ 2 seeds, 30-epoch frozen probe). Pool-mean Dice and trainable-parameter count per head design; the diagnostic's within-FM cross-head correlation summary is $\bar{r}$=0.690 (range 0.510–0.896), above the Table 1 RIGA Cup cross reference of 0.361.

Head design	Pool-mean Dice	Trainable params
`linear_1x1`	0.725	769
`mlp_2layer`	0.810	197,121
`unet_lite`	0.848	3,319,105
`transformer_decoder`	0.805	3,940,865

A33.2 Random-init CNN decorrelation

To check whether within-vs-cross structure persists in the absence of pretraining, we rerun the released random-init diagnostic on two CNN backbones (ConvNeXt-Tiny, ResNet-50) trained from scratch, 8 seeds $\{42-49\}$, on ACDC LV and RIGA Cup. Even with no shared pretraining, within-FM (cross-seed) $\bar{r}$ stays clearly above cross-FM (different-architecture) $\bar{r}$: 0.916 vs. 0.830 on ACDC LV (Δ=0.086), 0.841 vs. 0.720 on RIGA Cup (Δ=0.121). The gap is much smaller than the frozen-FM Table 1 numbers ($0.834/0.361$ on Cup, Δ=0.473), but it is still positive, consistent with architecture and shared recipe contributing to within-vs-cross structure in this diagnostic.

A33.3 Loss/augmentation sensitivity

The Section 4 protocol uses BCE loss, 224 inputs, no augmentation. To check whether the gap survives a closer-to-standard frozen-head loss/augmentation recipe, we evaluate the seven-bundle

Table 36: **Random-init CNN same/cross diagnostic.** Two CNN backbones $\times$ eight seeds; 30 epochs from scratch; same 1×1 probe protocol. Positive but smaller gaps suggest architecture/recipe dependence even without shared pretrained weights. Diagnostic only.

Task	within-FM $\bar{r}$	cross-FM $\bar{r}$	Δ
ACDC LV	0.916	0.830	0.086
RIGA Cup	0.841	0.720	0.121

subset {DINOv2-B/S, CLIP-L, ConvNeXt-T, EfficientNet-B0, ResNet-50/18} $\times$ 4 seeds with combined soft Dice+BCE loss and paired {hflip, rot$\pm15°$} augmentation. All runs still keep the encoder frozen, use the same 1×1-conv decoder, Adam 10^{-3}, and encoder-native 224×224 input; this is not a full clinical segmentation pipeline. The original RIGA Cup row remains the primary loss/augmentation diagnostic in Appendix A18; we also report three task-level diagnostic extensions (RIGA Disc, ACDC LV, and an ISIC-2018 Task 1 fixed 400-image held-out subset) to check whether the positive gap is an isolated Cup result. ISIC here is a supplementary per-case Dice diagnostic on a fixed 400-image subset and is not the official ISIC challenge test/server protocol or threshold-Jaccard leaderboard score.

Table 37: **Loss/augmentation sensitivity on RIGA Cup, 7 FMs $\times$ 4 seeds.** This is a closer-to-standard frozen-head recipe using Dice+BCE and simple augmentation, not a full clinical segmentation pipeline; all rows remain frozen 1×1 heads at 224×224. Per-FM seed-averaged Dice; pool $\bar{r}$ at the bottom. CLIP-L uses the same caveated OpenCLIP `ViT-L-14/openai` dense-token wrapper as the primary bundle.

FM	mean Dice	σ (seed)
DINOv2-B	0.758	0.005
DINOv2-S	0.733	0.003
CLIP-L	0.720	0.006
EfficientNet-B0	0.626	0.002
ResNet-50	0.618	0.010
ConvNeXt-T	0.583	0.004
ResNet-18	0.526	0.027
Pool: within $\bar{r}$=0.903, cross $\bar{r}$=0.556, Δ=0.347 [0.275, 0.428].		

Table 38: **Task-level loss/augmentation extensions.** Same seven-bundle, four-seed frozen-head Dice+BCE+augmentation recipe as Table 37. These rows are not primary rows. Dice columns summarise per-bundle seed-averaged Dice across the seven bundles. The ISIC row is a fixed 400-image-subset held-out diagnostic, not the official ISIC test/server evaluation.

Task	N_{test}	FM-mean Dice	FM min	FM max	Δ	95% CI
RIGA Cup	95	0.652	0.526	0.758	0.347	$[0.275, 0.428]$
RIGA Disc	95	0.853	0.756	0.922	0.617	$[0.535, 0.693]$
ACDC LV	361	0.565	0.389	0.740	0.326	$[0.291, 0.366]$
ISIC-2018 staged	80	0.864	0.801	0.918	0.473	$[0.375, 0.562]$

Across all four loss/augmentation rows, the Dice+BCE+augmentation recipe leaves a positive within-vs-cross gap (Δ = 0.326–0.617). The magnitude varies by task: ACDC remains lower accuracy and lower-gap than the RIGA/ISIC rows, while RIGA Disc shows the largest extension gap. These rows are treated as scope checks rather than primary evidence because they use a seven-FM subset and, for ISIC, a fixed subset/split; they indicate that the simplified BCE/no-augmentation protocol is not the sole source of the co-failure signal.

A33.4 Five-fold CV robustness

The fold-reslicing analysis is reported with the nnU-Net complements in Appendix A17 and Table 19. The key scope point is that the released artifact has one decoder seed per fold; it supports split-stability of cross-bundle $\bar{r}$, not a within-fold vs cross-fold same-backbone estimator.

A33.5 BraTS foreground train-size sweep

This train-size diagnostic covers the five BraTS foreground bundles with completed feature caches (DINOv2-B, BiomedCLIP, CLIP-L/14, EfficientNet-B0, ResNet-50). It uses one decoder seed (42) and fractions $\{0.10, 0.25, 0.50, 1.00\}$ of the same cached training set, so it is an *accuracy/saturation* check rather than a within-vs-cross dependence estimator. The result is non-monotone: the 5-bundle mean Dice rises from 0.579 at 10% to 0.621–0.622 at 25–50%, then stays flat/slightly lower at full data (0.617). Thus, in this cached five-bundle subset, adding more 2D FLAIR slices improves the weakest low-data setting but does not by itself create a new high-performing, decorrelated pool; the correlation claim still has to be audited directly rather than inferred from sample size.

Table 39: **BraTS train-size accuracy/saturation diagnostic, one seed only** (5 cached bundles, seed 42). Mean/min/max Dice are across bundles at each train fraction; this table does not estimate within/cross dependence because each bundle has one seed.

Train fraction	Cells	Mean Dice	Min Dice	Max Dice
0.10	5	0.579	0.418	0.721
0.25	5	0.621	0.455	0.750
0.50	5	0.622	0.436	0.751
1.00	5	0.617	0.458	0.737

A33.6 BraTS multimodal RGB derivative

This analysis checks whether the BraTS foreground co-failure signal is an artifact of using a unimodal FLAIR derivative. We build a deliberately simple three-channel 2D derivative with the same held-out subjects and top-10 foreground-area slice rule as Table 1, but pack three MRI modalities into RGB channels: R=T1c, G=T2w, B=T2-FLAIR. This is *not* the official 3D BraTS challenge protocol; it is a controlled frozen-feature stress test of the input-channel choice.

The evaluated grid contains 20 cells (5 FMs $\times$ 4 seeds). Multimodal input narrows the dependence gap relative to the unimodal Table 1 BraTS row, but does not remove it: case-level within $\bar{r}$=0.904, cross $\bar{r}$=0.654, Δ=0.249 [0.238, 0.262]. Aggregating each subject's slices before correlation gives within $\bar{r}$=0.946, cross $\bar{r}$=0.772, Δ=0.174. The conservative interpretation is therefore narrower than "BraTS input modality does not matter": adding T1c/T2w raises cross-family agreement and shrinks the gap, yet same-backbone/family dependence remains positive under the same frozen-head audit.

Table 40: **Simple three-channel BraTS 2D derivative** (R=T1c, G=T2w, B=T2-FLAIR; 5 FMs $\times$ 4 seeds). This is the same 2D frozen-head audit, not the official 3D BraTS multimodal challenge protocol. Dice is seed-averaged by FM; dependence uses the same Pearson estimator as Table 1.

FM	Mean Dice	Min Dice	Max Dice	Seeds
BiomedCLIP	0.713	0.669	0.739	4
DINOv2-B	0.703	0.675	0.721	4
DINOv2-S	0.702	0.690	0.714	4
CLIP-L/14	0.686	0.682	0.692	4
ResNet-50	0.461	0.439	0.473	4
Case-level pool: within $\bar{r}$=0.904, cross $\bar{r}$=0.654, Δ=0.249 [0.238, 0.262].				
Subject-aggregated pool: within $\bar{r}$=0.946, cross $\bar{r}$=0.772, Δ=0.174.				

A33.7 MC dropout as a within-family decorrelator

A natural pool-design question is whether MC dropout [Gal and Ghahramani, 2016] can act as a lightweight within-family decorrelation lever. We rerun DINOv2-B on RIGA Cup with three dropout rates $p \in \{0.1, 0.2, 0.4\}$, 4 seeds, 10 MC samples per case. The deterministic within-FM cross-seed $\bar{r}\approx$0.97–0.99; under MC averaging, $\bar{r}$ drops to 0.95 (p=0.1), 0.93 (p=0.2), 0.89 (p=0.4). The largest tested drop is ≈ 0.10 and remains materially smaller than the same-architecture-family ViT-vs-primary-cross separation seen on Cup (Table 3). MC dropout is therefore a weak within-family lever in this diagnostic; consistent with Appendix A13, it does not approach the architecture-family separation.

Table 41: **MC-dropout diagnostic.** DINOv2-B/RIGA Cup with four seeds and 10 MC samples per case; correlations are computed on MC-averaged per-case Dice. MC dropout provides modest decorrelation but remains far above the Table 1 RIGA Cup cross-family reference of 0.361. Diagnostic only.

Dropout p	MC mean Dice	Det. within $\bar{r}$	MC within $\bar{r}$	H_{pred}
0.1	0.666	0.972	0.947	0.062
0.2	0.657	0.972	0.925	0.067
0.4	0.633	0.987	0.893	0.076

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim that *in the frozen and light-adaptation regime of medical segmentation pipelines*, same-FM/same-recipe pools show higher per-case Dice-score correlation, used as a failure-dependence proxy, than the Table-1 cross-family reference pools. The primary table uses task-specific functional pools after Dice-floor filtering: Kvasir 11 FMs, ACDC LV 10, BraTS foreground 10, RIGA Cup 9, and RIGA Disc 11. The scope is explicitly restricted to retrospective public-benchmark dependence audits; secondary pixel/lift readouts, released nnU-Net complements, and non-primary mechanism diagnostics are diagnostic or scope context rather than causal clinical-outcome evidence.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The main paper gives high-level limitations in Section 8, and the appendix expands them with scoped protocol notes: frozen/light-adaptation only; LoRA and RIGA full-fine-tuning probes without distributed traces retained only as transparency notes; released CNN random-init and small ACDC full-fine-tuning provenance traces are diagnostic, not primary evidence; UNet-with-skip and task-level nnU-Net v2 complements reported but not fully case-identical dense 3D FM/full-pipeline baselines for every primary row; the 3D architecture pilot is appendix scope context only; 2D axial slices at 224×224 for the primary FM audit; MedSAM used only as a frozen vision-encoder probe; broader cross-scanner / cross-demographic shifts remain untested; clinical Dice thresholds are operational conventions and many lift thresholds are support-limited.

3. **Theory assumptions and proofs**

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4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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   Justification: Section 4, Appendix A13, and Appendix A32 specify the public FM checkpoints, decoder architecture (1×1-conv default plus stated sweeps), training protocol (Adam 10^{-3}, 30 epochs for the frozen-head primary runs, batch 16, BCE for the binary primary rows), seed set ($\{42, 43, 44, 45\}$ with σ-seeded determinism), and the actual primary split protocols: Kvasir 880/120, RIGA train = BinRushed+MESSIDOR with Magrabia (Magrabi Eye Center) source held out (n=95), ACDC public patients 001–100 with patient-level fold-0 LV-positive slices (n=361), and the BraTS 2024 2D FLAIR `seg>0` foreground derivative: per held-out subject, up to the top-10 axial slices by foreground area with at least 64 positive pixels, subject-level 80/20 split using NumPy seed 42 (n=3228). This BraTS target includes the GLI resection-cavity label and is not the official 3D BraTS WT challenge target. The supplementary artifact releases primary-task per-case Dice JSONs, merged summaries, selected aggregate diagnostic JSONs, code, command documentation, hashes, and a data card for trace-level reproduction; raw-data retraining requires separately obtaining upstream datasets/checkpoints, and full raw data, full prediction caches, raw held-out prediction arrays, and fuller split manifests are outside the supplementary artifact.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 gives the formal estimands (within-/cross-family $\bar{r}$, spatial Δ_{pix}, threshold lift L) and the symmetric 4×4 seed-combo averaging rule; Section 4 gives the datasets (with patient-level splits), foundation-model list, and protocol (image size, encoder-specific upstream normalisation, decoder, optimiser, loss, epochs, batch, seed control).

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 1 reports paired case-level and hierarchical family/seed bootstrap 95% CIs (B=2000). The BraTS foreground derivative is analysed at the slice level with an explicit within-subject dependence caveat, Appendix A11 reports subject-level sensitivity checks, and Appendix A12 explains why sparse-class NCR filtering diagnostics are not used as numeric evidence.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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Justification: Appendix A32 reports the compute/resource scope. For trace-level verification, the artifact is directly reproducible on CPU in minutes through the artifact verification script. Raw-image reproduction used NVIDIA A100-80GB, A6000, and L40S GPUs across a shared cluster. Feature extraction per FM per domain ranges from minutes for CNN backbones to longer ViT/BraTS jobs; cached 1×1-head training per seed is lightweight. For raw-image retraining, we report resource classes and per-job elapsed times where recorded, but do not claim a single exact total GPU-hour count because raw retraining is outside the submitted artifact's reproduction scope.

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